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Antidiabetic effects of polyherbal mixture made of *Centaurium erythraea*, *Cichorium intybus* and *Potentilla erecta*

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A R T I C L E I N F O

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A B S T R A C T

Ethnopharmacological relevance: The polyherbal mixture made of *Centaurium erythraea* aerial parts and *Cichorium intybus* roots and *Potentilla erecta* rhizomes has been used for centuries to treat both the primary and secondary complications of diabetes.
Aim of the study: As a continuation of our search for the most effective herbal mixture used as an ethnopharmacological remedy for diabetes, this study aimed to compare the *in vitro* biological activities of this polyherbal mixture and its individual ingredients, and, most importantly, to validate the ethnopharmacological value of the herbal mixture through evaluation of its phytochemical composition, its potential *in vivo* toxicity and its effect on diabetes complications.
Materials and methods: Phytochemical analysis was performed using HPLC-UV. Antioxidant activity was estimated via the DPPH test. Potential cytotoxicity/anticytotoxicity was assessed using an *in vitro* RBCs antihemolytic assay and an *in vivo* sub-chronic oral toxicity method. Antidiabetic activity was evaluated using an *in vitro* α-amylase inhibition assay and *in vivo* using a chemically induced diabetic rat model.
Results: The HPLC-UV analysis revealed the presence of *p*-hydroxybenzoic acid, *p*-hydroxybenzoic acid derivative, catechin, five catechin derivatives, epicatechin, isoquercetin, hyperoside, rutin, four quercetin derivatives, caffeic acid, and four caffeic acid derivatives in the polyherbal mixture decoction. Treatment with the decoction has shown no toxic effects. The antioxidant and cytoprotective activities of the polyherbal mixture were higher than the reference's ones. Its antidiabetic activity was high in both *in vitro* and *in vivo* studies. Fourteen days of treatment with the decoction (15 g/kg) completely normalized blood glucose levels of diabetic animals, while

Abbreviations: AKT, protein kinase B; AGE, Advanced glycation end product; ALP, Alkaline phosphatase; ALT, Alanine transaminase; ANOVA, One-way analysis of variance; AST, Aspartate transaminase; Bcl-2, B-cell lymphoma 2; BDNF, Brain-derived neurotrophic factor; BH, Binucleated hepatocytes; BHT, Butylated hydroxytoluene; CA1, *Cornu ammonis* 1; CA2, *Cornu ammonis* 2; CA3, *Cornu ammonis* 3; CA4, *Cornu ammonis* 4; COX-2, Cyclooxigenase-2; CREB, cAMP response element-binding protein; D-10, Diabetic animal group treated with the polyherbal mixture (10 g of dry plant material/kg); D-15, Diabetic animal group treated with the polyherbal mixture (15 g of dry plant material/kg); D-2.5, Diabetic animal group treated with the polyherbal mixture (2.5 g of dry plant material/kg); D-5, Diabetic animal group treated with the polyherbal mixture (5 g of dry plant material/kg); D-C, Diabetic animal control group; D-G, Diabetic animal group treated with glimepiride; D-I, Diabetic animal group treated with insulin; DMSO, Dimethyl sulfoxide; DNSA, 3,5-dinitrosalicylic acid; H–C, Healthy control group; DPPH, 2,2-Diphenyl-1-picrylhydrazyl; ERK1/2, Extracellular signal-regulated kinase 1/2; GAE, Gallic acid anhydrous; TPC, Total phenolic content; H-10, Healthy animal group treated with the polyherbal mixture (10 g of dry plant material/kg); H-15, Healthy animal group treated with the polyherbal mixture (15 g of dry plant material/kg); H-2.5, Healthy animal group treated with the polyherbal mixture (2.5 g of dry plant material/kg); H-5, Healthy animal group treated with the polyherbal mixture (5 g of dry plant material/kg); HDL, High-density lipoprotein; H&E, Hematoxylin and eosin; HMG-CoA, 3-hydroxy-3-methylglutaryl-coenzyme A; CYP2E1, cytochrome P450 2E1; IDF, International Diabetes Federation; IL-1β, Interleukin-1β; IL-6, Interleukin-6; iNOS, Inducible nitric oxide synthase; JNK, c-Jun N-terminal kinase; LDL, Low-density lipoprotein; MDA, Malondialdehyde; MH, Mononucleated hepatocytes; MAPK, mitogen-activated protein kinase; MMP, matrix metalloproteinase; mTOR, mammalian target of rapamycin; NAFLD, Non-alcoholic fatty liver disease; N:C, nucleus:citoplasm; ND-C, Non-diabetic animal control group; NF-κB, Nuclear factor kappa B; NPC1L1, NPC1 like intracellular cholesterol transporter 1; Nrf2, Nuclear factor erythroid 2-related factor 2; PAS, Periodic acid-Schiff; PKA, Protein kinase A; PT, Proximal tubule; QuE, Quercetin hydrate; RBCs, Red blood cells; PBS, Phosphate buffered saline; ROS, Reactive oxygen species; SGLT, sodium-glucose linked transporter; TBA, Thiobarbituric acid; TCA, Trichloroacetic acid; TFC, Total flavonoid content; TIMP-1, tissue inhibitors of metalloproteinase 1; TLR4, Toll-like receptor 4; TNF-α, Tumor necrosis factor-α; TRIS, 2-amino-2-(hydroxymethyl)propane-1,3-diol; VLDL, Very-low-density lipoprotein.

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treatments with insulin and glimepiride only slightly lowered glycemic values. In addition, lipid status of treated animals as well as levels of serum AST, ALT, ALP, creatinine, urea and MDA were completely normalized. In addition, the polyherbal mixture completely restored the histopathological changes of the liver, kidneys and all four *Cornu ammonis* regions of the hippocampus.

Conclusions: The polyherbal mixture was effective in the prevention of both primary and secondary diabetic complications such as hyperlipidemia, increased lipid peroxidation, non-alcoholic fatty liver disease, nephropathy and neurodegeneration.

1. Introduction

Diabetes mellitus is a complex metabolic disorder where chronic hyperglycemia over time often leads to the progression of a number of secondary complications (Bastaki, 2005; Parvin et al., 2014), such as hyperlipidemia, non-alcoholic fatty liver disease (NAFLD), nephropathy (Evans et al., 2002; Guven et al., 2006; Savage et al., 2007; Manna et al., 2010; Palsamy et al., 2010), neuropathy, even dementia and the risk of Alzheimer's disease (Ott et al., 1999; Arvanitakis et al., 2004; Inzucchi et al., 2015).

If not treated in a timely manner, these secondary complications affect the quality of life of diabetics and may also lead to the death of the patient (Papatheodorou et al., 2018). According to the International Diabetes Federation (IDF), there were over 1.2 million children and adolescents, 537 million adults living with diabetes, and as many as 6.7 million deaths due to diabetes in 2021. Considering that in Africa, Southeast Asia, and the Western Pacific, more than half of people with diabetes are undiagnosed, these numbers might be even higher (IDF, 2021).

Even though standard diabetes pharmacotherapies successfully regulate hyperglycemia, their use is often accompanied by numerous side effects, such as weight gain caused by insulin therapy (Russell-Jones and Khan, 2007), genitourinary candidiasis and decrease in bone mass caused by SGLT inhibitors (Dowarah and Singh, 2020), headache and anorexia caused by pramlintide acetate (Dowarah and Singh, 2020), gastrointestinal complications caused by metformin (Kirpichnikov et al., 2002) and increased risk of myocardial infarction and weight gain caused by sulfonylureas (Panten et al., 1996; Schloot et al., 2016). Also, these pharmacotherapies do not fully address the problem of elevated levels of AGE and ROS, the main causes of progression of secondary diabetic complications (Amparo et al., 2017). Thus, many contemporary studies focus on the development of new herbal supplements capable of preventing secondary complications of diabetes, and the use of traditional medicine has been flourishing (Seven et al., 2004; Pan et al., 2016; Li et al., 2021). At the same time, a trend of declining or stable incidence of diabetic comorbidities has been reported in more than 80% of the world since 2010 (IDF, 2021).

Yet, the potential toxicity and complex composition limit the rapid development and application of herbal medicines, and both compound analysis and evaluation of clinical efficacy are essential (Kim and Ha, 2013; Brown, 2017; Yang et al., 2018). Thus, in every preliminary pre-clinical study, three major steps are a must – compound characterization, *in vitro* study, and *in vivo* evaluation.

Qualitative and quantitative analysis of bioactive compounds is usually performed using high-performance liquid chromatography (HPLC) (Fang et al., 2015; Shuo et al., 2016; Jiafu Feng et al., 2016). Afterwards, it is necessary to clarify whether their chemical profiles support traditional medicinal values. The first step is *in vitro* evaluation of biological aspects such as antioxidant, antidiabetic, and cytoprotective potential, as well as the degree of toxicity. However, *in vivo* evaluation using an appropriate approach, such as a chemically induced diabetic animal model, is crucial. At this stage, the ethnopharmacological value of the treatment is evaluated by analyzing biochemical parameters such as MDA, AST, ALT, urea, creatinine, blood glucose, lipid status, as well as through histopathological analysis, where some of the most convenient histochemical staining such as hematoxylin and eosin, together with detailed morphometric analysis, can provide deep insights into the therapeutic properties of medical herbs (Wang et al., 2009; Wang et al., 2012; Yi-Na Tang et al., 2015; Madić et al., 2021).

Our previous studies have shown that one of the polyherbal mixtures used as a remedy for diabetes in European countries improves hyperglycemia and has hypolipidemic, nephroprotective, hepatoprotective (Madić et al., 2021), and osteoprotective (Petrović et al., 2021) effects. However, according to our previous research (Madić et al., 2018), direct conversations with numerous ethnopharmacologists, as well as literature data (Životić and Životić, 1979), there are other different herbal mixtures that are commonly used, especially by people in the Balkans, which could be even more effective as antidiabetic agents.

One of them consists of the aerial parts of common centaury (*Centaurium erythraea* Rafin., Gentianaceae), the roots of common chicory (*Cichorium intybus* L., Asteraceae) and rhizomes of tormentil (*Potentilla erecta* (L.) Räuschel, Rosaceae) and has been used for a long time as a treatment for diabetes accompanied with its secondary complications; taken after the meal for at least seven consecutive days (Životić and Životić, 1979).

These medical herbs are well known for their antioxidant, hypoglycemic, hypolipidemic, anti-inflammatory, antimicrobial, hepatoprotective, antiflatulent, gastroprotective, antitumor, and nephroprotective activities (Wiater et al., 2008; Ahmed, 2009; Daraz et al., 2010; Zlatković et al., 2013; Liu et al., 2013; Yao et al., 2013; Dalar and Konczak, 2014; Gospodinova and Krasteva, 2015; Trifunović-Momčilović et al., 2016; Gholami et al., 2018; Mihaylova et al., 2019; Kachmar et al., 2019; Bouyahya et al., 2019; Zeb et al., 2019; Đorđević et al., 2020). However, many studies have shown that when addressing multifactorial diseases such as diabetes and its secondary complications, treatments with polyherbal preparations may be more beneficial than the use of single medicinal plants (Mahajan et al., 2018; Madić et al., 2021; Huang et al., 2022; Chen et al., 2023).

Having in mind all mentioned above, and the fact that despite the common use of this polyherbal preparation as a remedy for the treatment of both primary and secondary diabetic complications, there is no research data to validate its traditional value, this study aimed to compare *in vitro* biological activities of aqueous extracts of the polyherbal mixture with the activities of its individual ingredients, as well as to validate the ethnopharmacological use of the polyherbal mixture through evaluation of its phytochemical composition, potential *in vivo* toxic effects and, above all, its possible hypoglycemic, hypolipidemic, hepatoprotective, nephroprotective, and neuroprotective effects.

2. Materials and methods

2.1. Chemicals

Butylated hydroxytoluene ≥99% (BCBW5813), 2,2-Diphenyl-1-picrylhydrazyl (STBG8201), Quercetin hydrate ≥95% (STBH7170), Malondialdehyde tetrabutylammonium salt ≥96% (0000205459), D-(+)-maltose monohydrate ≥99% (M5885), Potato starch (S5651), Pancreatic α-amylase (A3176, type VI-B), Rutin hydrate ≥94% (BCBH0339V), (+)-Catechin hydrate 98% (BCBF0735V), *p*-Coumaric acid ≥98% (BCCG8417) were purchased from Sigma Aldrich (USA). Caffeic acid 99.5% (30304) was obtained from Dr. Ehrenstorfer GmbH

(Germany). Gallic acid monohydrate, 98% (L830T36) was provided by J&K scientific GmbH (Germany). 3,5-Dinistrosalicylic acid ≥97% (10221149) was obtained from Alfa Aesar (USA). Dimethyl sulfoxide (504341000) was purchased from Biochem Chemopharma (France). Alloxan-monohydrate 98% (A0421158) was provided from from Acros Organics (Belgium). Gallic acid anhydrous (S7588849835) was obtained from Merck (USA). Folin-Ciocalteu's reagent (T123085I) was provided by Carlo Erba Reagents (Spain). Ketamidor 10% (0921733AA) was purchased from Richter pharma (Austria). Insulin glargine (2F8395A) and glimepiride (U634) were provided by Sanofi (France). Total cholesterol ((10)66478201), High-density lipoprotein ((10)65300701), Triglycerides ((10)69959801), Aspartate transaminase ((10)67010301), Alanine transaminase ((10)67320501), Alkaline phosphatase ((10) 664840), Creatinine ((10)664770401), and Urea ((10)70152501) assay kits were purchased from Roche Diagnostics GmbH, (Deutschland). All the other chemicals used were analytical-grade reagent chemicals.

2.2. Plant material and extract preparation

Examined plant species were collected in South-Eastern Serbia during 2020. *C. intybus* and *C. erythrea* were collected at Stara planina Mt (43 21.943 N; 22 46.000 E), and *P. erecta* at Vlasina Lake Plateau (42 43.940 N; 22 19.533 E). The plants were taxonomically identified, and voucher specimens were deposited at the herbarium collection of the Faculty of Science and Mathematics, University of Niš, under the accession numbers HMN 14454–14460.

Plant material was dried in dark conditions at room temperature for 3 weeks. *C. intybus* roots, *C. erythrea* aerial parts, and *P. erecta* rhizomes (1:1:5, respectively) were mixed to prepare the polyherbal mixture according to the traditionally used recipe (Životić and Životić, 1979). The aqueous extracts were prepared as a decoction. 100 g of each tested plant material and the polyherbal mixture were boiled separately in 1000 mL of distilled water until half of the liquid evaporated. The obtained aqueous extracts were filtered, and the solvent was totally removed using a vacuum evaporator (IKA RV10 Rotary Evaporator with HB10 Bath, Gaithersburg). Extraction yield was calculated using the formula (Zhang et al., 2018)

$$Yield\ (\%) = \frac{Weight\ of\ the\ dry\ extract\ (g)}{Weight\ of\ the\ dry\ plant\ (g)} \times 100$$

2.3. Phytochemical analysis

2.3.1. Determination of total phenolic and total flavonoid content

The total phenolic content (TPC) of extracts was determined spectrophotometrically by the Folin-Ciocalteu method as previously described (Madić et al., 2021). 0.3 mL of tested extract (1 mg/mL, diluted in methanol) was mixed with 1.5 mL Folin-Ciocalteu's reagent (diluted in methanol 1:10 (v/v)) and 1.2 mL of Na_2CO_3 (7.5%). After 2h of incubation time in the dark at room temperature, the absorbance was measured at 740 nm (UV–1650PC, Shimadzu 1650, Europe). The total phenolic concentration was calculated from a gallic acid (GAE) calibration curve (10–100 mg/L, diluted in methanol). Data were expressed as gallic acid equivalents (GAE)/g of the extract.

The total flavonoid content (TFC) was determined as previously described (Madić et al., 2021). Briefly, 1 mL of tested extract (1 mg/mL, diluted in methanol) was mixed with 6.4 mL of distilled water (dH_2O), 0.3 mL of $NaNO_2$ (5%), 0.3 mL of $AlCl_3$ (10%) and 2 mL of NaOH (1 M) and incubated in the dark at room temperature for 30 min. The total flavonoid content in extracts was calculated from a quercetin hydrate (QuE) calibration curve (10–100 mg/L, diluted in methanol) at 510 nm and expressed as quercetin equivalents (QuE)/g of dry extract.

2.3.2. HPLC analysis

Individual phytochemical compounds of the polyherbal mixture decoction were evaluated using HPLC-UV detection on the Agilent 1200 Series with the C18 column (Zorbax Eclipse XDB C18, 5 μm, 4.6 × 150 mm), Diode Array Detector (DAD), quaternary pump, vacuum degasser, autosampler and AgilentChemStation software (Agilent Technologies, US), as previously described (Madić et al., 2021). Briefly, 2 mL of decoction was filtrated using a microfilter. Elution of the samples was done in gradient mode, by varying the volume ratios (v/v) of eluent A (0.27 M formic acid solution) and eluent B (methanol): initially 70% A (0–5 min), 70-30% A (5–20 min), 30-10% A (20–25 min). The injected sample volume was 5 μL, the flow was 1 mL/min, the temperature of the column was 25 °C, the wavelength range was 190–400 nm, and the detection wavelengths were 254, 280, and 350 nm. Identification of the extract components was done by comparing retention times and UV absorption spectra of constituents with commercially available standards. Quantification of identified compounds was performed based on calibration curves of gallic acid (gallic acid, *p*-hydroxybenzoic acid, and derivatives), catechin (catechin and derivatives), caffeic acid (caffeic acid and derivatives), *p*-coumaric acid (*p*-coumaric acid and derivates, ferulic acid derivates), rutin (rutin and other glycosides).

2.4. In vitro study

2.4.1. DPPH radical scavenging assay

The DPPH radical scavenging assay was used to evaluate the antioxidant activity of tested extracts (Blois, 1958). The dried aqueous extracts were dissolved in 100% methanol. 0.3 mL of different concentrations of each extract, i.e., 5, 10, 15, 25, 50, 75, 100 μg/mL were mixed with 2.7 mL of DPPH radical methanol solution (0.04 mg/mL), and incubated in the dark at room temperature for 30 min. The absorbance was measured at 517 nm. 0.3 ml of methanol with DPPH radical methanol solution was taken as a positive control. The radical scavenging activity (DPPHsa) was calculated by the formula (Madić et al., 2021):

$$DPPHsa\ (\%) = [(Ap - Ax) / Ap]\ x100$$

where Ap was the absorbance of the positive control, and Ax absorbance of the tested extract.

Butylated hydroxytoluene (BHT) was used as a standard.

2.4.2. α-amylase inhibition assay

The *α*-amylase inhibition assay was done by a slightly modified method of Ali et al. (2006). Potato starch (1% w/v) was diluted in 0.02 M sodium phosphate buffer (enriched with 6.7 mmol/L sodium chloride, pH 6.9) and stabilized by heating at 70 °C for 15 min. Five concentrations of each extract, i.e., 10, 25, 50, 75, and 100 μg/mL were prepared by diluting the stock solution in dimethyl sulfoxide (DMSO). 1 mL of each extract was mixed with 1 mL of α-amylase (1U/mL). After 20 min of preincubation period at 37 °C, 1 mL of the preincubated mixture was mixed with 1 mL of potato starch and incubated at 37 °C for 20 min 1 mL of DNSA color reagent was added to each tested sample, boiled for 5 min, cooled down, and diluted in 9 mL of dH_2O. In the end, the generation of maltose was calculated from a maltose calibration curve at the wavelength of 540 nm.

α -amylase inhibition activity was calculated as follows(Bhandari et al., 2008):

$$Inhibition\ (\%) = [(Ac - As / Ac]x100$$

where Ac was the absorbance of the positive control, and As absorbance of the tested extract.

2.4.3. RBCs antihemolytic assay

Red blood cells (RBCs) were prepared as previously described (Madić et al., 2019). In short, Wistar rats were anesthetized with 10% Ketamidor and the blood sample was collected in heparinized tubes via cardiopuncture following the principles of the Care and Use of Laboratory Animals and approved by the Ethical Committee of the Faculty of

Medicine, University of Niš (No. 323-07-08987/2022-05). The whole blood was centrifuged at 2000 rpm for 10 min at 4 °C. Plasma and the buffy coat were removed, and an equal volume of PBS (0.9%, pH 7.4) was added. This was repeated three times to obtain washed RBCs. The washed RBCs were diluted in PBS buffer to obtain a 4% suspension and used immediately.

The dried aqueous extracts were dissolved in PBS. 0.5 mL of different concentrations of each tested extract, i.e., 25, 50, 75, 125, and 250 μg/mL were gently mixed with 1 mL of RBCs suspension and with enough PBS to reach the final volume of 2.5 mL. After 5 min of incubation at room temperature, 0.25 mL of 0.3% H_2O_2 was added to induce oxidative degradation and the mixture was incubated at 37 °C for 4 h. At the end of the incubation period, the samples were centrifuged at 2000 rpm for 5 min at 4 °C. 0.2 mL of the obtained supernatant was mixed with 3 mL of Drabkin's reagent, incubated for 10 min at room temperature and, in the end, the absorbance was measured at 540 nm. The negative control was RBCs in PBS, and the positive control was RBCs in the presence of H_2O_2 and the absence of the tested plant extracts. The antihemolytic activity was calculated as follows(Madić et al., 2019):

$$Antihemolytic\ activity\ (\%) = [(Apc - As) / (Apc - Anc)]x100$$

where Apc was the absorbance of the positive control, As absorbance of the tested extract, and Anc absorbance of the negative control. Butylated hydroxytoluene (BHT) was used as a standard.

2.5. In vivo study

2.5.1. Experimental animals

To assess the effect the of the polyherbal mixture on primary and secondary complications of diabetes, we used an alloxan-induced diabetic rat model (Gupta et al., 2020; Madić et al., 2021) because treatment with alloxan monohydrate induces both primary and secondary diabetic complications. Namely, administration of alloxan causes damage and death of pancreatic β-cells (Rohilla and Ali, 2012) and a resulting increase in blood glucose levels. Moreover, this treatment also leads to the development of a number of pathological changes in other organs such as the liver and kidneys (Elangovan et al., 2019; Ganesan et al., 2020; Madić et al., 2021) and even damage and loss of neurons in the brain (Šimić et al., 1997; Middha et al., 2012). Because chemically induced diabetes might be unstable this study was performed in female rats (Vital et al., 2006) and included only animals with blood glucose levels above 20 mmol/L for 14 consecutive days after alloxan injection, i.e., animals with stable diabetes (Madić et al., 2021).

Female Wistar rats (220–270 g) were obtained from the Institute of Biomedical Research, Medical Faculty, Niš, Serbia. Animals were kept under standard husbandry conditions, with a 12/12 h light/dark cycle, a temperature of 23 ± 2 °C, relative humidity of 55 ± 10%, fed with standard rat food pellets and water provided *ad libitum*; following the principles of the Care and Use of Laboratory Animals and approved by the Ethical Committee of the Faculty of Medicine, the University of Niš (No. 323-07-08987/2022-05).

2.5.2. Experimental design of sub-chronic toxicity study

A sub-chronic toxicity study was done in accordance with OECD 407 (Repeated Dose 28-day (OECD, 2018), with some modifications (Madić et al., 2021). 25 animals were randomly divided into 5 groups of 5 animals. Four groups (H-2.5, H-5, H-10, and H-15) were treated with tested polyherbal mixture decoction (2.5, 5, 10, and 15 g of dry plant material/kg, respectively), and the fifth group, i.e., healthy controls (H–C) was treated with water, by oral gavage, for the 28 days. The doses of evaluated polyherbal mixture decoction varied from the minimal to the maximal ones used in ethnopharmacology (Životić and Životić, 1979) multiplied by 6 (Nair and Jacob, 2016). The bodyweight was recorded daily, while the blood glucose level was measured weekly from blood collected from the tail vein using a portable glucometer (FreeStyle Precision Neo, Abbott Diabetes Care Inc., UK).

2.5.3. Experimental design of the anti-diabetic study

Of 40 animals used in this study, diabetes was induced in 35 of them by intraperitoneal injection of alloxan monohydrate (150 mg/kg) dissolved in ice-cold PBS, after overnight fasting, while the remaining 5 animals were treated only with PBS and used as a healthy controls (non-diabetic control group – ND-C). To prevent hypoglycemia, 1 h after the alloxan injection, the rats had been given a 5% glucose solution over the next 48 h, *ad libitum* (Madić et al., 2021)

Animals with blood sugar higher than 20 mmol/L fourteen days after alloxan administration were considered diabetic and randomly divided into seven groups of five animals each (D-C, D-2.5, D-5, D-10, D-15, D-I, and D-G). For the next two weeks, D-2.5, D-5, D-10, and D-15 groups were orally treated with Polyherbal mixture extract (2.5, 5, 10, and 15 g of dry plant material/kg, respectively); D-G group orally with glimepiride (1 mg/kg); while D-I group received insulin glargine intraperitoneally (13 IU/kg). Diabetic controls (D-C groups) and non-diabetic controls (ND-C group) received only water by oral gavage. The bodyweight was recorded daily. The blood glucose level was measured before the alloxan injection (day 0), fourteen days after the alloxan injection (day 14), and at the end of the study (day 28), using a portable glucometer (FreeStyle Precision Neo, Abbott Diabetes Care Inc., UK).

2.5.4. Tissue sampling

At the end of the experiment, after overnight fasting, animals were anesthetized with 10% Ketamidor. Blood samples were collected by cardiopuncture and centrifugated at 3500 rpm for 10 min at 4 °C to obtain serum for the subsequent biochemical analyses. Pancreas, liver, and left kidney samples were washed in PBS and fixed in 10% formalin, while brain samples were fixed in 4% formalin. Right kidneys of experimental animals were used for the evaluation of malondialdehyde (MDA) level.

2.5.5. Biochemical analysis

Total cholesterol, high-density lipoprotein (HDL), and triglycerides for lipid profile analysis, as well as aspartate transaminase (AST), alanine transaminase (ALT), alkaline phosphatase (ALP), creatinine, and urea, were measured spectrophotometrically on automated chemistry analyzer (Cobas c311, Roche) according to the manufacturer's instructions (Roche Diagnostics gmbH, Deutschland). The level of very low-density lipoprotein (VLDL) was determined as the concentration of triglycerides/5. The concentration of low-density lipoprotein (LDL) was calculated using the formula: (Friedewald et al., 1972).

$$LDL(mmol / L) = triglycerides - HDL - VLDL$$

2.5.6. Lipid peroxidation assay

Lipid peroxidation evaluation was done using the slightly modified method of Ohkawa and associates (1979). Briefly, 10% homogenates (1:10, w/v) of kidney tissues in TRIS-HCl (0.5 M, pH 7.4) were centrifugated at 10000 rpm for 10 min at 4 °C. 2 mL of the obtained supernatant as well as 2 mL of serum was separately mixed with 2 mL TBA/TCA/HCl reagent and incubated at 96 °C for 30 min, cooled, and centrifugated at 4000 rpm for 10 min at 4 °C. At the end, the concentration of malondialdehyde was calculated from the MDA calibration curve at the wavelength of 532 nm and expressed as nmol/g of kidney tissue and μmol/mL of serum.

2.5.7. Histopathological analysis

The liver, kidney, brain and pancreatic tissue were embedded in paraplast and cut into 5 μm thick sections (Leica RM2125 RT, Germany). The hepatic, renal and pancreatic tissues were subjected to hematoxylin and eosin (H&E) and Masson's trichrome staining, while the liver and kidneys were additionally analyzed using the periodic acid–Schiff (PAS) method. Brain tissue sections were subjected to Nissl staining. Five to ten sections per sample were imaged using a Leica microscope (Leica DM2500, Ger many) at 200x and 400x magnification, respectively, and analyzed by ImageJ software (NIH, USA) (Anjomshoa et al., 2020; Madić et al., 2021)

2.6. Statistical analysis

Statistical analysis was done using GraphPad Prism 5 (GraphPad Software, La Jolla California USA). *In vitro* experiments were done in triplicate, *in vivo* experiments in pentaplicate, and data were expressed as the mean ± standard deviation. The differences between the controls and the individual dosage groups of the tested extract were analyzed by the one-way analysis of variance (ANOVA) followed by Tukey's Multiple Comparison Test. The relation between total phenol, total flavonoid content, IC_{50} of DPPH scavenging activity, and IC_{50} of the α-amylase inhibition potential of tested extracts was assessed using Pearson's correlation coefficient. Statistical differences were accepted if p was less than 0.05. In the figures, '*' denotes statistical significance where there is only one control group in the experiment, 'a' denotes statistical significance compared to the untreated diabetic controls, 'b' – statistical significance compared to the healthy controls, 'c'- statistical significance compared to the diabetic animals treated with insulin, and 'd' - statistical significance compared to the diabetic animals treated with glimepiride.

3. Results and discussion

3.1. Phytochemical analysis

The total phenol content of all tested extracts was positively correlated with their total flavonoid content ($p < 0.05$, r = 0.97), as shown in Table 1. The polyherbal mixture had significantly higher amount of phytochemical components (TPC: 198.33 ± 1.53 mg of GAE/g TPC, TFC: 88.67 ± 8.5 mg of QuE/g) than two of its ingredients, i.e., *C. intybus* (TPC: 93.67 ± 1.52 mg of GAE/g, TFC: 45.33 ± 8.39 mg of QuE/g) and *C. erithrea* (TPC: 39.33 ± 0.57 mg of GAE/g, TFC: 25.33 ± 1.53 mg of QuE/g) ($p < 0.05$). At the same time, *P. erecta* extract had even higher total phenolic and flavonoid content (TPC: 266.67 ± 2.52 mg of GAE/g, TFC: 145.68 ± 8.71 mg of QuE/g) than polyherbal mixture one ($p < 0.05$) (Table 1).

HPLC-UV analysis revealed the presence of 21 phytochemical compounds in the polyherbal mixture extract: catechin, catechin derivatives, epicatechin, *p*-hydroxybenzoic acid, *p*-hydroxybenzoic acid derivative, caffeic acid, caffeic acid derivatives, quercetin 3-*O*-galactoside, quercetin 3-*O*-glucoside, quercetin 3-*O*-rutinoside, and other quercetin derivatives, as shown in Fig. 1.

The most abundant components were *p*-hydroxybenzoic acid (810 μg/mL), catechin (764 μg/mL), catechin derivatives 3, 1, 2, 4 and 5 (650.07, 408.71, 321.07, 203.49 and 174.40 μg/mL, respectively), *p*-hydroxybenzoic acid derivative (170.78 μg/mL), epicatechin (146.21 μg/mL), quercetin derivative 4, 3 and 2 (92.42, 79.87, 79.84 and 58.87 μg/mL, respectively), and quercetin 3-*O*-glucoside (30.84 μg/mL).

Table 1
Yield, total phenolic content (TPC) and total flavonoid content (TFC) of tested extracts.

Extract	Extraction yield (%)	TPC (mg of GAE/g of extract)	TFC (mg of QuE/g of extract)
polyherbal mixture	18.88	198.33 ± 1.53	88.67 ± 8.50
C. erythrea	51.34	39.33 ± 0.57*	25.33 ± 1.53*
C. intybus	21.69	93.67 ± 1.52*	45.33 ± 8.39*
P. erecta	24.14	266.67 ± 2.52*	145.68 ± 8.71*

Data were expressed as the mean ± standard deviation (n = 3), *$p < 0.05$ compared to the polyherbal mixture.

Amounts of the remaining identified metabolites were quercetin 3-*O*-rutinoside (19.05 μg/mL), caffeic acid (17.79 μg/mL), caffeic acid derivatives 3, 2, 1 and 4 (14.02, 12. 22, 10.73 and 10.46 μg/mL, respectively), quercetin 3-*O*- galactoside (11.53 μg/mL) and quercetin derivative 1 (7.95 μg/mL) (Table 2).

3.2. In vitro study

3.2.1. DPPH radical scavenging activity

Comparative *in vitro* analysis using DPPH test showed that extracts of the polyherbal mixture, *C. erythrea* and *P. erecta* showed higher antioxidant activity than BHT (p < 0.05), used as a standard. Moreover, the extract of the polyherbal mixture exhibited significantly higher antioxidant activity than two of the ingredients, *C. erythrea* and *C. intybus* (Fig. 2 (A)),

The observed antioxidant activity can be explained by the higher content of phenols and flavonoids (Table 1), due to the higher content of *P. erecta* than *C. erythrea* and *C. intybus* used in the traditionally used recipe (5:1:1). This is in agreement with previous studies that showed that herbal mixtures have higher antioxidant and cytoprotective activities than most of their individual ingredients when used alone (Madić et al., 2019; Mapeka et al., 2022).

3.2.1.1. α-amylase inhibition potential. Similarly to the results of the DPPH test, the α-amylase inhibitory activity of all tested extracts was dose-dependent ($p < 0.001$), as shown in Fig. 2 (B). However, its enzyme inhibition activity was similar to that of *P. erecta* and much higher than that of *C. intybus* and *C. erythrea* extracts ($p < 0.05$), medicinal herbs already known to inhibit digestive enzymes (Uysal et al., 2018; Bouyahya et al., 2019; Menyiy et al., 2021). This is an important aspect in the application of diets enriched with polyphenols because it is possible to prevent hyperglycemia (Zhang et al., 2018) and stabilize blood glucose in diabetic patients by inhibiting this enzyme (Ali et al., 2006; Demir et al., 2019; Ganesan and Xu., 2019).

3.2.2. RBCs antihemolytic activity

In the RBCs antihemolytic assay, the modulatory treatments with all the tested extracts, except *P. erecta*, showed a strong inhibitory effect against H_2O_2-induced hemolysis of rat erythrocytes at some of the tested concentrations. Interestingly, as shown in Fig. 3, the lowest tested concentration of the polyherbal mixture (25 μg/mL) had a stronger cytoprotective activity than the same concentration of its individual ingredients, i.e., the polyherbal mixture (93.01 ± 0%) > *C. intybus* (72.53 ± 3.43%) > *C. erythrea* (59.05 ± 1.91%) > *P. erecta* (7.176 ± 0%). Except for the extract of *P. erecta*, which showed cytotoxic activity at almost all the tested concentrations, all the other extracts had higher antihemolytic activity than BHT ($p < 0.05$).

These results are in agreement with previous studies in which natural polyphenols generally showed stronger antioxidant and cytoprotective capacity than synthetic ones such as BHT (von Gadow et al., 1997; Sebranek et al., 2005; Yehye et al., 2015). In addition, many secondary metabolites such as flavonoids can protect erythrocyte membranes from oxidative degradation and act preventively against the consequences of constant hyperglycemia (Sharma and Sharma, 2001). The low cytoprotective activity of the aqueous extract of *P. erecta*, even the cytotoxic one (Fig. 3), is attributable to the well-known phenomenon called the 'Janus effect'. Namely, bioactive substances can act both as radical scavengers and radical producers, depending on their concentration (Zeiger, 2003; Madić et al., 2019).

Considering that the high content of polyunsaturated fatty acids and oxygen transport associated with redox-active hemoglobin molecules make red blood cells (RBCs) the first targets of reactive oxygen species (ROS) that can lead to secondary complications caused by ischemia (Rice-Evans et al., 1986), the results of *in vitro* evaluation encouraged us to pursue to the *in vivo* evaluation of polyherbal mixture decoction.

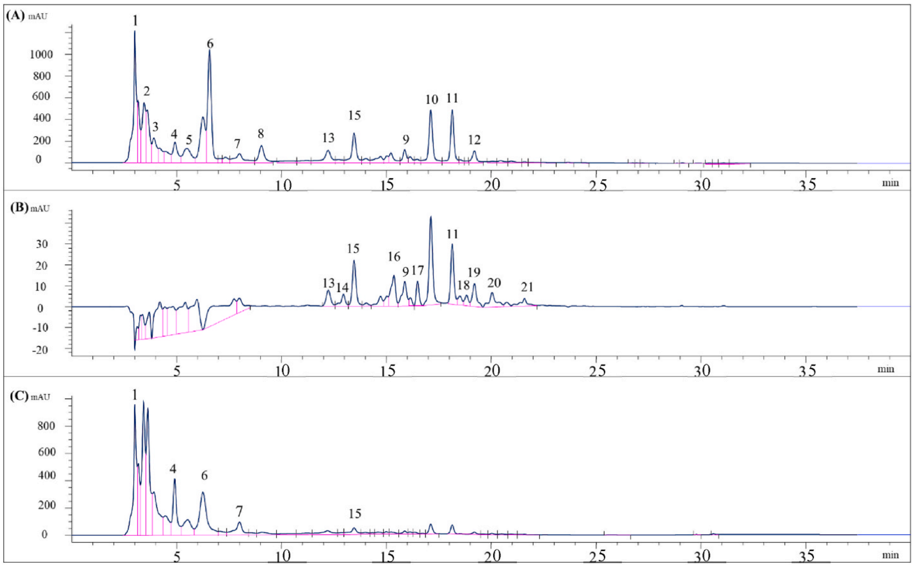

Fig. 1. HPLC chromatogram of the polyherbal mixture decoction recorded at 254 (A), 350 (B), and 280 nm (C). 1: catechin; 2: catechin derivative-1; 3: catechin derivative-2; 4: catechin derivative-3; 5: catechin derivative-4; 6: catechin derivative-5; 7: *p*-hydroxybenzoic acid; 8: epicatechin; 9: *p*-hydroxybenzoic acid derivative; 10: quercetin derivative-1; 11: caffeic acid derivative-1; 12: quercetin derivative-2; 13: quercetin derivative-3; 14: caffeic acid; 15: quercetin 3-*O*-galactoside; 16: caffeic acid derivative-2; 17: quercetin 3-*O*-glucoside; 18: quercetin 3-*O*-rutinoside; 19: quercetin derivative-4; 20: caffeic acid derivative-3; 21: caffeic acid derivative-4.

Table 2
Characterization and concentration (μg/mL) of phytochemical compounds in polyherbal mixture decoction identified by HPLC-UV.

	Compound	R.t. (min)	Conc. (μg/mL)
1	catechin	3.015	746.00
2	catechin derivative 1	3.455	408.71
3	catechin derivative 2	3.609	321.07
4	catechin derivative 3	3.635	650.07
5	catechin derivative 4	3.915	203.49
6	catechin derivative 5	4.929	174.40
7	*p*-hydroxybenzoic acid	6.582	810.88
8	epicatechin	8.015	146.21
9	*p*-hydroxybenzoic acid derivative	9.055	170.78
10	quercetin derivative 1	12.249	7.95
11	caffeic acid derivative 1	12.969	10.73
12	quercetin derivative 2	13.475	58.87
13	quercetine derivative 3	14.735	79.84
14	caffeic acid	15.375	17.79
15	quercetin 3-*O*-galactoside (hyperoside)	15.895	11.53
16	caffeic acid derivative 2	16.502	12.22
17	quercetin 3-*O*-glucoside (isoquercetine)	17.135	30.84
18	quercetin 3-*O*-rutinoside (rutin)	18.155	19.05
19	quercetin derivative 4	19.215	92.42
20	caffeic acid derivative 3	20.055	14.02
21	caffeic acid derivative 4	21.602	10.46

R.t.: retention time; P.A.: Peak area; Conc.: Compound concentration in decoction.

3.3. *In vivo study*

3.3.1. *Sub-chronic toxicity study*

3.3.1.1. The effect of the polyherbal mixture on healthy animals' blood glucose level and body weight. Treatment with the polyherbal mixture caused no change in glycemic values for the first three weeks. However, we observed a slight decrease in blood glucose level of all the treated animals compared to the untreated ones on day 28 (Fig. 4). Yet, even decreased, the blood sugar levels of treated animals were within normal range (Wang et al., 2010).

Moreover, by the end of the experiment, treatment with the polyherbal mixture caused a slight decrease in body weight gain of treated animals compared to the untreated controls (Fig. 5), which is consistent with previous data showing that a polyphenol-enriched diet reduces adipogenesis and lipogenesis (Zhao et al., 2017) through targeting multiple signaling pathways and regulation of the mammalian target of rapamycin (mTOR) signaling pathway (Cao et al., 2022).

3.3.1.2. The effect of the polyherbal mixture on healthy animals' lipid profile and biochemical parameters. Twenty-eight days of treatment with the polyherbal mixture did not result in any changes in the lipid status of the treated healthy animals (Supplementary Fig. 1) or in the creatinine, urea, and MDA levels (Fig. 6 (E, F, D), Supplementary fig. 4 (H)) compared to the untreated animals, indicating normal function of liver and kidneys (Ozer et al., 2008; Gowda et al., 2010.) and the absence of increased oxidative stress (Oktem et al., 2005; Tsikas, 2017).

However, all tested concentrations of the extract showed a tendency

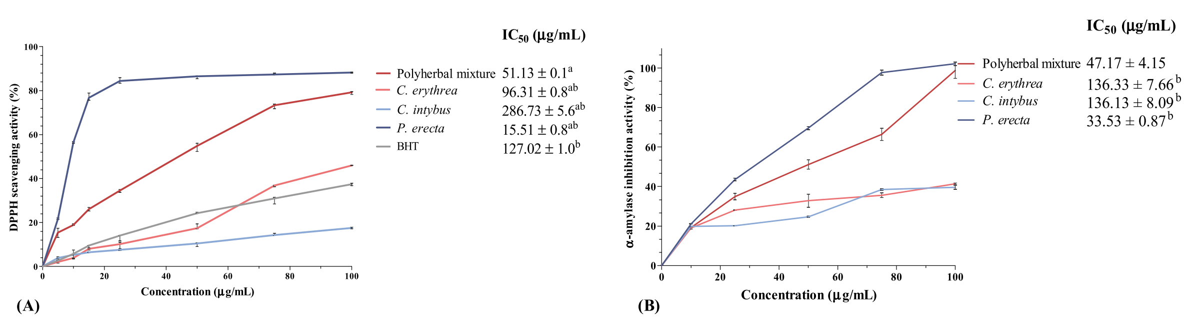

Fig. 2. DPPH scavenging activity (A), α-amylase inhibition activity (B). Data were expressed as the mean ± standard deviation (n = 3), $^{a}p < 0.05$ compared to the BHT; $^{b}p < 0.001$ compared to the polyherbal mixture.

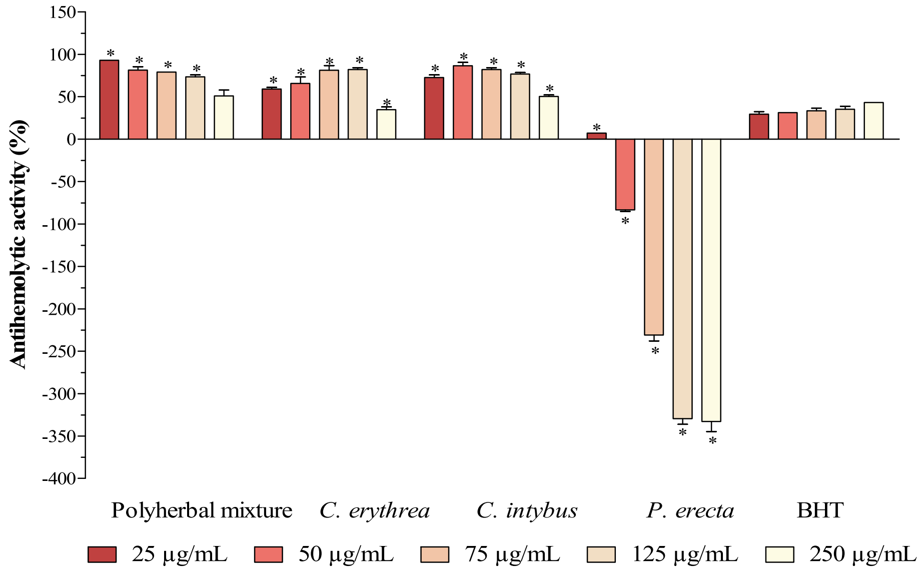

Fig. 3. Antihemolytic activity of tested extracts. BHT: butylhydroxytoluene. Data were expressed as the mean ± standard deviation (n = 3), *$p < 0.05$ compared to the BHT.

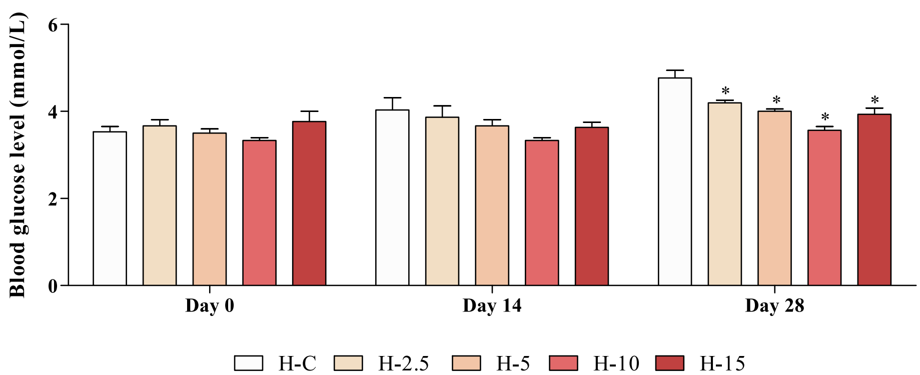

Fig. 4. Blood glucose level of the experimental groups on days 0, 14 and 28 in the sub-chronic toxicity study. H–C: Healthy control; H-2.5: Polyherbal mixture 2.5 g/kg; H-5: Polyherbal mixture 5 g/kg; H-10: Polyherbal mixture 10 g/kg; H-15: Polyherbal mixture 15 g/kg. Data were expressed as the mean ± standard deviation (n = 5), *$p < 0.05$ compared to the H–C group.

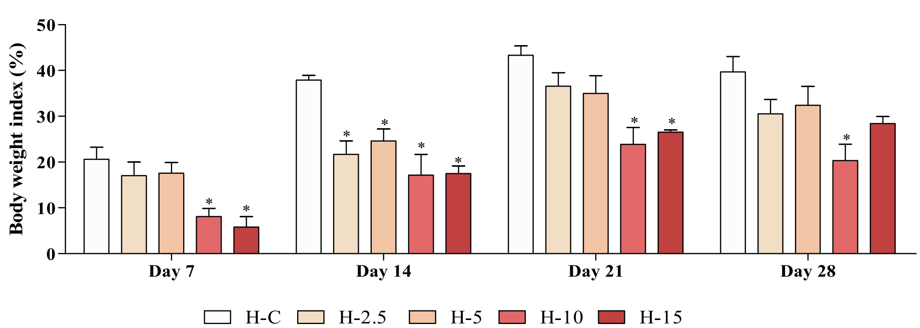

Fig. 5. The percentage of the body weight index change of experimental groups on days 7, 14, 21 and 28 in the sub-chronic toxicity study. H–C: Healthy control; H-2.5: Polyherbal mixture 2.5 g/kg; H-5: Polyherbal mixture 5 g/kg; H-10: Polyherbal mixture 10 g/kg; H-15: Polyherbal mixture 15 g/kg. Data were expressed as the mean ± standard deviation (n = 5), *$p < 0.05$ compared to the H–C group.

to decrease the levels of AST and ALT (Fig. 6 (A, B)), whereas they significantly decreased the level of ALP (Fig. 6 (C)) of the treated animals compared to the untreated controls ($p < 0.05$), suggesting the hepatoprotective abilities of the tested polyherbal mixture (Zhang et al., 2018; Oršolić et al., 2021).

3.3.1.3. The effect of the polyherbal mixture on healthy animals' liver, kidney, and hippocampal histology. Histopathological analysis revealed that treatment with any of the tested concentrations of the polyherbal mixture did not cause structural changes in the pancreas (Supplementary Fig. 2), liver (Supplementary Fig. 3), kidney (Supplementary Fig. 4), or hippocampus (Supplementary Fig. 5) of the experimental animals, which confirmed the absence of hepatotoxicity (Popescu et al., 2012;

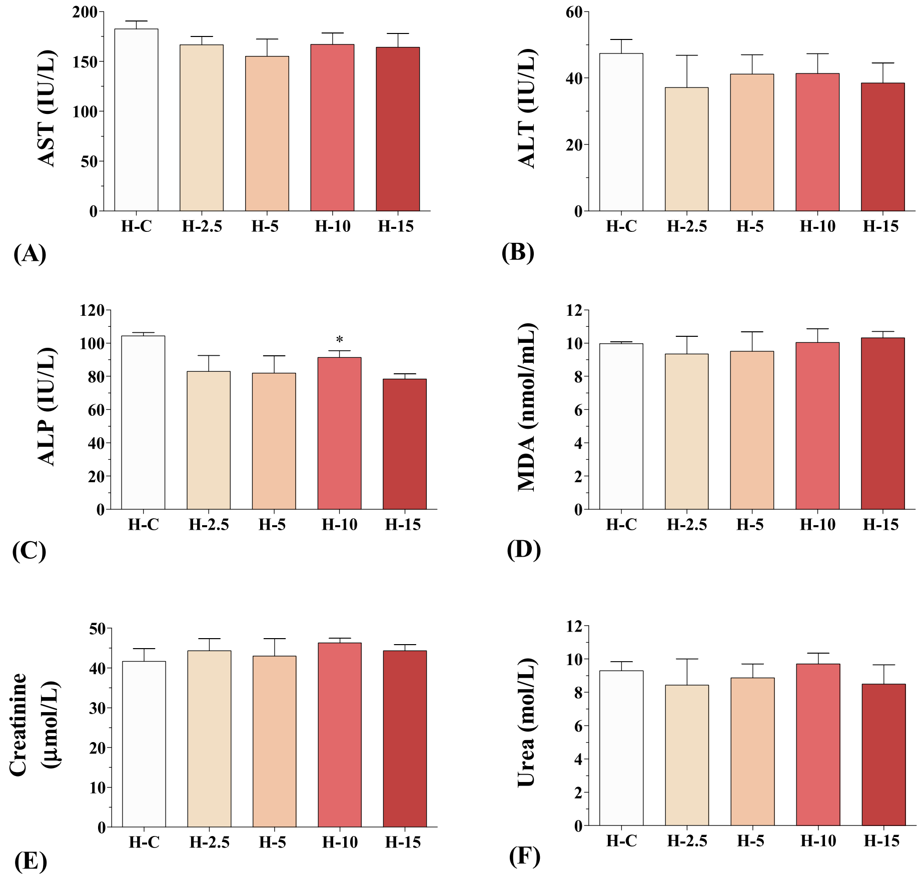

Fig. 6. Serum AST (A), ALT (B), ALP (C), MDA (D), creatinine (E) and urea (F) level of experimental groups in the sub-chronic toxicity study. H–C: Healthy control; H-2.5: Polyherbal mixture 2.5 g/kg; H-5: Polyherbal mixture 5 g/kg; H-10: Polyherbal mixture 10 g/kg; H-15: Polyherbal mixture 15 g/kg. Data were expressed as the mean ± standard deviation (n = 5), *$p < 0.05$ compared to the H–C group.

Wilkinson et al., 2019; Karsdal et al., 2020), nephrotoxicity (Alsaad and Herzenberg, 2007; Baues et al., 2020; Bouteldja et al., 2021) and neurotoxicity (Hori et al., 2010; Gong et al., 2017; Omotoso, 2018; Morroni et al., 2018) of the tested polyherbal mixture decoction.

Moreover, considering that treatment with the herbal mixture did not induce any changes in the number of β-cells (Supplementary Fig. 2 (C)) and that there were no disturbances in the metabolism of carbohydrates seen as unchanged deposition of liver glycogen (Supplementary Fig. 3 (J)), we might conclude that the observed slight decrease of blood glucose levels of treated animals compared to the untreated controls (Fig. 4) is due to the high α-amylase inhibitory activity of the polyherbal mixture extract (Fig. 2 (B)).

3.3.2. *Antidiabetic effect study*

3.3.2.1. The effect of the polyherbal mixture on diabetic animals' blood glucose level and body weight. As shown in Fig. 7, two weeks after the injection of alloxan monohydrate, blood glucose levels were significantly higher in all diabetic groups compared to the healthy controls ($p < 0.001$). By the end of the experiment, treatment with 10 g/kg of the herbal mixture significantly lowered glycemic values (8.77 ± 0.11 mmol/L) compared to the diabetic controls (23.87 ± 1.05 mmol/L) ($p < 0.001$), whereas treatment with the highest tested concentration (15 g/kg) completely restored blood glucose level to normoglycemic values (5.24 ± 0.06 and 4.53 ± 0.15 mmol/L in experimental groups D-15 and ND-C, respectively). Treatments with insulin and glimepiride also decreased the blood glucose (15.72 ± 0.84 and 17.47 ± 0.75 mmol/L, respectively) compared to the untreated diabetic animals ($p < 0.001$), however, to a much lesser extent than higher doses of the polyherbal

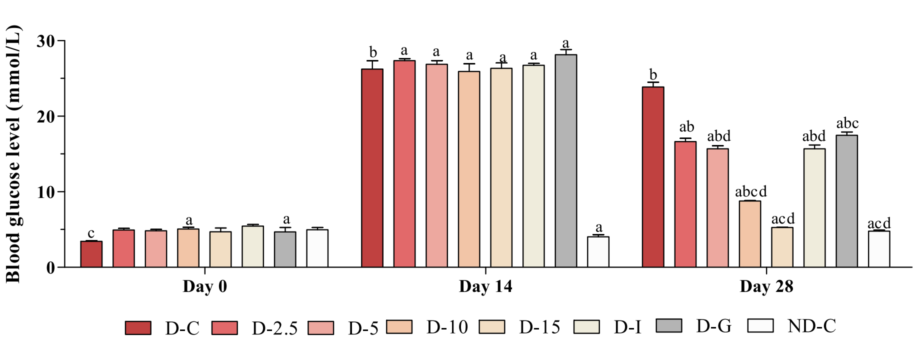

Fig. 7. Blood glucose level of experimental groups on days 0, 14 and 28 of the anti-diabetic study. D–C: Diabetic control; D-2.5: Polyherbal mixture 2.5 g/kg; D-5: Polyherbal mixture 5 g/kg; D-10: Polyherbal mixture 10 g/kg; D-15: Polyherbal mixture 15 g/kg; D–I: Insulin; D–G: Glimepiride; ND-C: non-diabetic control. Data were expressed as the mean ± standard deviation (n = 5), $^{a}p < 0.001$ compared to the D-C group; $^{b}p < 0.001$ compared to the ND-C group; $^{c}p < 0.001$ compared to the D-I group; $^{d}p < 0.001$ compared to the D-G group.

mixture ($p < 0.001$) (Fig. 7).

These results are in concordance with the study of Latha and Daisy (2011) and our previous research (Madić et al., 2021) that showed that potent extracts of medicinal plants and herbal mixture may have higher hypoglycemic activity than treatment with insulin in a rat model of diabetes. Moreover, having in mind that glimepiride is the standard drug for type 2 diabetes in a period where there are still at least some functional beta-cells, the observed low efficiency in the prevention of hyperglycemia is additional proof that induction of late-stage type 2 diabetes in animals was successful (Panten et al., 1996).

Moreover, histopathological analysis of pancreatic tissue showed that the number of β-cells in the islets of Langerhans was significantly decreased (Supplementary Fig. 6 (C)) while the deposition of collagen in the pancreas was significantly increased (Supplementary Fig. 6 (D)), which is additional evidence for the stability of diabetes until the end of the experiment (Madić et al., 2021). The treatments with higher tested concentrations of the polyherbal mixture (10 and 15 g/kg) fully restored the number of β-cells in the islets of Langerhans (Supplementary Fig. 6 (C)) and managed to decrease the deposition of collagen in pancreatic tissue as well (Supplementary Fig. 6 (D)).

As presented in Fig. 8, after confirmation of stable diabetes, we observed significant body loss in diabetic controls (-11.73 ± 4.80 and $-16.91 \pm 5.87\%$ on days 7 and 14, respectively) compared to the healthy ones (22.68 ± 6.81 and $38.51 \pm 9.67\%$ on days 7 and 14, respectively) ($p < 0.001$). Of all the treatments, the most profound ameliorative effect on weight loss caused by diabetes was shown by treatment with 15 g/kg of the herbal mixture (20.68 ± 12.64 and $26.22 \pm 13.47\%$ on days 7 and 14, respectively) and the lowest one by treatment with glimepiride (9.39 ± 8.79 and $17.25 \pm 13.65\%$ on days 7 and 14, respectively).

This is consistent with literary data where unregulated diabetes, as well as chemically induced diabetes, results in weight loss due to the initiation of gluconeogenesis from other energy sources such as lipids (Kim and Ha, 2013; Yan et al., 2016).

Additionally, these results agree with previous studies in which hydroxybenzoic acid, catechin, quercetin, isoquercetin, hyperoside, caffeic acid, and rutin, highly abundant bioactive compounds in this polyherbal mixture decoction (Table 2), showed hypoglycemic activities in rats with type 1 diabetes (Peungvich et al., 1998; Jadhav and Puchchakayala, 2012; Pitchai and Manikkam, 2012; Jayachandran et al., 2018; Lee et al., 2020; Oršolić et al., 2021). Moreover, it has already been observed that concomitant administration of catechin, epicatechin, and rutin exhibits higher antihyperglycemic activity than derivatives of sulfonylureas (Ahmed et al., 2012; Mechchate et al., 2021). Namely, many polyphenols decrease the level of pro-apoptotic Bax protein level and downsize the apoptosis of beta cells (Gurzov and Eizirik, 2011; Arya et al., 2014), which helps beta cells to regenerate.

3.3.2.2. The effect of the polyherbal mixture on diabetic animals' lipid profile. At the end of the experiment, the levels of total cholesterol, LDL, triglycerides and VLDL were significantly increased in the diabetic controls compared to the healthy ones ($p < 0.001$), whereas HDL levels were significantly decreased in the D-C group compared to the H–C group ($p < 0.001$), as presented in Fig. 9. Treatment with the polyherbal mixture (15 g/kg) restored all biochemical parameters of lipid status to the level of the healthy control group, i.e., it significantly lowered total cholesterol, LDL, triglycerides and VLDL, and increased HDL levels.

This was expected because is known that hypoinsulinemia leads to an imbalance of lipid metabolism in adipocytes, which promotes the accumulation of lipids in blood vessels and the potential progression of atherosclerosis (Ito et al., 2019). The observed hypolipidemic activity of herbal mixture is due to the presence of hydroxybenzoic acid, isoquercetin, hyperoside, caffeic acid, and rutin in its decoction (Table 2, Fig. 1). Namely, as many other polyphenols and flavonoids, these bioactive compounds are already known for their hypolipidemic and hypoglycemic activities (Celik et al., 2009; Zhang et al., 2018; Ganesan et al., 2020; Aldayel et al., 2020; Oršolić et al., 2021), achieved by inhibition of NPC1 like intracellular cholesterol transporter 1 (NPC1L1) (Kobayashi, 2019) and by attenuation of cholesterol synthesis in the liver via competitive inhibition of 3-hydroxy-3-methylglutaryl-coenzyme A (HMG-CoA) reductase (Islam et al., 2015).

3.3.2.3. The effect of the polyherbal mixture on diabetic animals' biochemical parameters. Serum levels of AST, ALT, ALP, creatinine, and urea were significantly increased in diabetic controls compared to the healthy controls ($p < 0.001$). Two weeks of treatment with the highest tested concentration of the herbal mixture (15 g/kg) completely normalized all these parameters (Fig. 10). At the same time, treatment with insulin and glimepiride slightly lowered these biochemical parameters but could not bring them to the level of healthy animals.

Animals in the diabetic control group had significantly higher serum creatinine and urea levels compared to the healthy animals (Fig. 10 (E, F)), indicating renal dysfunction, one of the most common secondary complications in both type 1 and type 2 diabetes. To wit, impairments of the renal tubular epithelium and glomeruli lead to disruption of the urine filtration process and reabsorption of minerals and water, ultimately resulting in their imbalance in the organism (Chang et al., 2005; Oktem et al., 2005; Gowda et al., 2010). Treatment with higher tested doses of the polyherbal mixture (10 and 15 g/kg) significantly reduced alloxan-induced diabetic nephropathy in terms of decreasing the serum creatinine and urea levels (Fig. 10).

In addition, malondialdehyde levels were significantly elevated in both serum and kidney tissue in the untreated diabetic animals (6.28 ± 1.22 mmol/mL of serum and 33.84 ± 1.50 nmol/g of renal tissue) compared to the healthy ones (9.98 ± 0.09 mmol/mL of serum and 13.53 ± 0.28 nmol/g kidney of renal tissue) ($p < 0.001$) (Figs. 10 and 12 (H)) indicating a lipid oxidation increase (Herawati et al., 2020). Treatment with the with polyherbal mixture significantly decreased malondialdehyde levels compared to diabetic controls in a concentration-dependent manner ($p < 0.001$), whereas 15 g/kg of polyherbal mixture completely restored these parameters to the level of healthy animals (10.86 ± 0.058 mmol/mL of serum and 14.95 ± 0.69 nmol/g of kidney tissue).

Since alloxan monohydrate is a toxic compound, the observed the

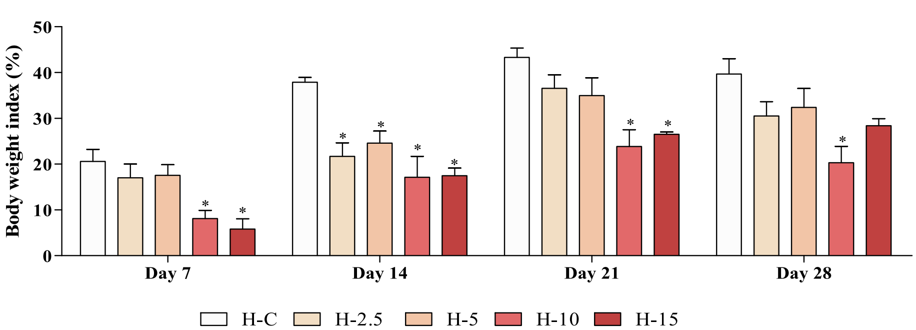

Fig. 8. The percentage of the body index change of experimental groups on days 7 and 14 after the beginning of treatment with the polyherbal mixture in the anti-diabetic study. D–C: Diabetic control; D-2.5: Polyherbal mixture 2.5 g/kg; D-5: Polyherbal mixture 5 g/kg; D-10: Polyherbal mixture 10 g/kg; D-15: Polyherbal mixture 15 g/kg; D–I: Insulin; D–G: glimepiride; ND-C: Non-diabetic control. Data were expressed as the mean ± standard deviation (n = 5), $^{a}p < 0.001$ compared to the D-C group; $^{b}p < 0.001$ compared to the ND-C group; $^{c}p < 0.001$ compared to the D-I group; $^{d}p < 0.001$ compared to the D-G group.

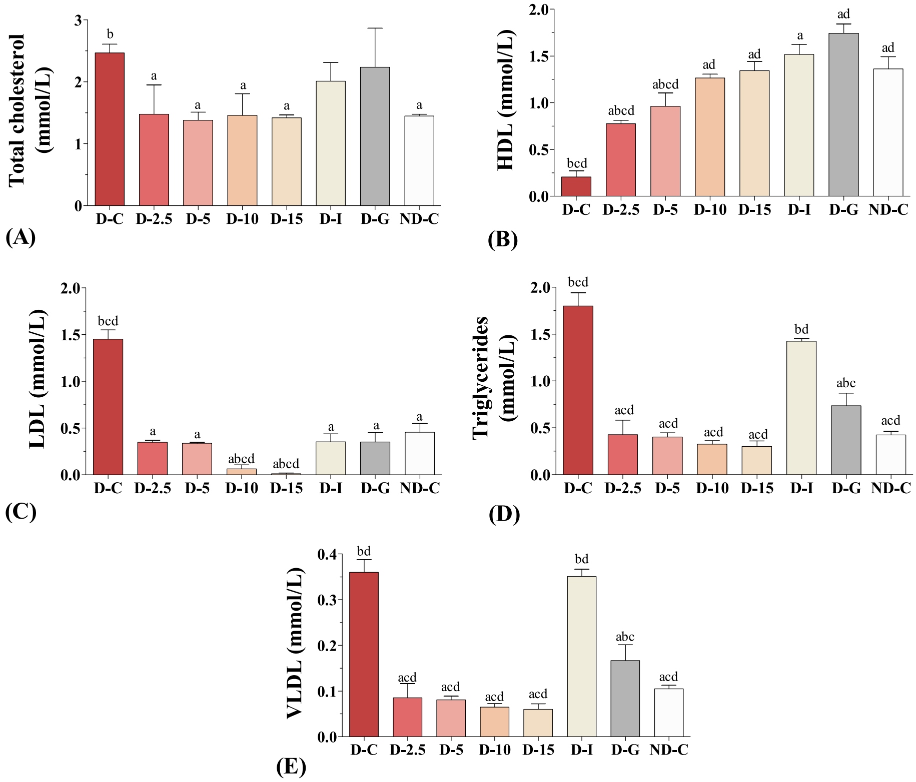

Fig. 9. Lipid profile of experimental groups in the anti-diabetic study. D–C: Diabetic control; D-2.5: Polyherbal mixture 2.5 g/kg; D-5: Polyherbal mixture 5 g/kg; D-10: Polyherbal mixture 10 g/kg; D-15: Polyherbal mixture 15 g/kg; D–I: Insulin; D–G: Glimepiride; ND-C: non-diabetic control. Data were expressed as the mean ± standard deviation (n = 5), $^{a}p < 0.001$ compared to the D-C group; $^{b}p < 0.001$ compared to the ND-C group; $^{c}p < 0.001$ compared to the D-I group; $^{d}p < 0.001$ compared to the D-G group.

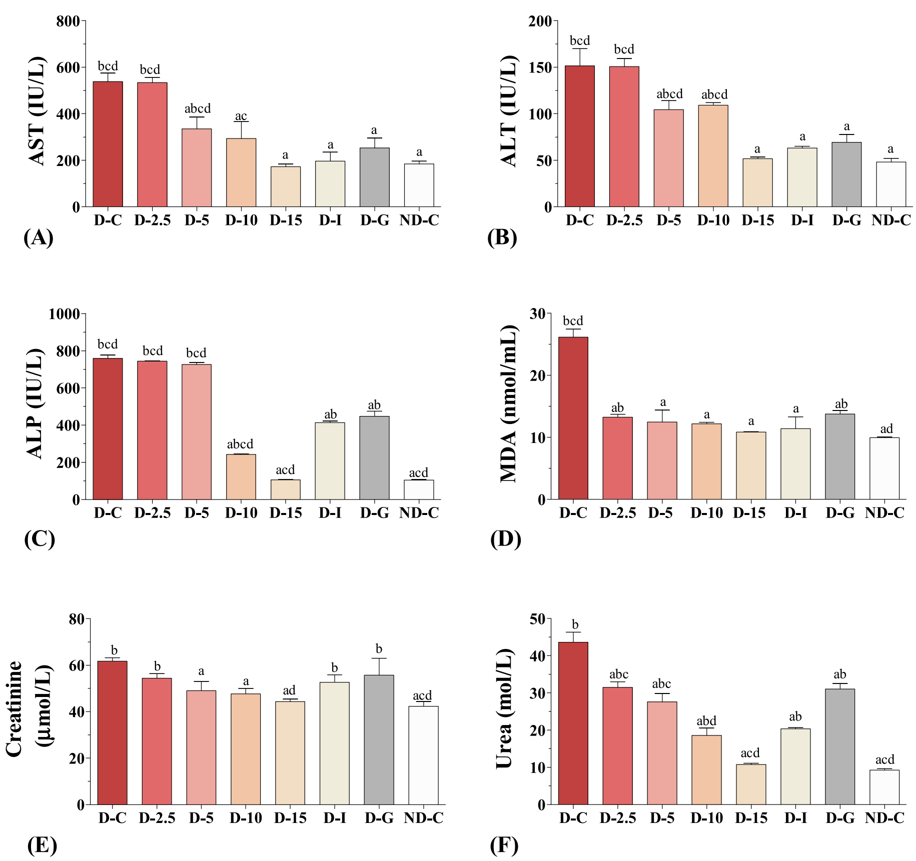

Fig. 10. Serum AST (A), ALT (B), ALP (C), MDA (D), creatinine (E) and urea (F) level of experimental groups in the anti-diabetic study. D–C: Diabetic control; D-2.5: Polyherbal mixture 2.5 g/kg; D-5: Polyherbal mixture 5 g/kg; D-10: Polyherbal mixture 10 g/kg; D-15: Polyherbal mixture 15 g/kg; D–I: Insulin; D–G: Glimepiride; ND-C: non-diabetic control; AST: aspartate transaminase; ALT: alanine transaminase; ALP: alkaline phosphatase. Data were expressed as the mean ± standard deviation (n = 5), $^{a}p < 0.001$ compared to the D-C group; $^{b}p < 0.001$ compared to the ND-C group; $^{c}p < 0.001$ compared to the D-I group; $^{d}p < 0.001$ compared to the D-G group.

liver response to high intoxication, caused by the initiation of the process of metabolizing the toxin, which subsequently led to liver tissue damage due to the state of oxidative stress (Lucchesi et al., 2013). Having in mind the high content of polyphenols in the polyherbal mixture decoction, in particular isoquercetin, hyperoside, and caffeic acid (Table 2, Fig. 1), the ameliorative effect of treatment are in agreement with previous studies. Namely, these compounds are already known for their hepatoprotective activities and ability to lower serum transaminases, ALP and MDA (Zhang et al., 2018; Oršolić et al., 2021), even more successfully than glimepiride (Schaalan et al., 2009; Aldayel et al., 2020).

3.3.2.4. The effect of the polyherbal mixture on diabetic animals' liver histology. Detailed morphometric analysis confirmed that treatment with polyherbal mixture had significant hepatoprotective capabilities. Namely, alloxan-induced diabetes caused deterioration of the liver tissue in terms of decreasing the percent of total hepatocytes, the percent of binucleated hepatocytes, and the ratio of nucleus:cytoplasm in both mononucleated and binucleated hepatocytes (60.95 ± 2.58, 8.36 ± 0.65, 15.97 ± 1.28, and 17.06 ± 0.64%, respectively) compared to the healthy controls (76.46 ± 2.14, 20.91 ± 2.44, 22.67 ± 1.44, and 28.04 ± 0.95%, respectively) ($p < 0.001$) (Fig. 11 (D, H, F, G)), which indicates the reduction of the regenerative capacity of the liver (Ozbek et al., 2017; Madić et al., 2021).

At the same time, the percent of Kupffer cells, critical mediators of NAFLD, was significantly increased in diabetic controls (39.05 ± 2.58%) compared to the healthy ones (25.97 ± 4.48%) ($p < 0.001$) (Fig. 11 (E)), suggesting an enhanced immune response of liver tissue (Karsdal et al., 2020; Madić et al., 2021).

Moreover, at the end of the experiment, the distribution of collagen

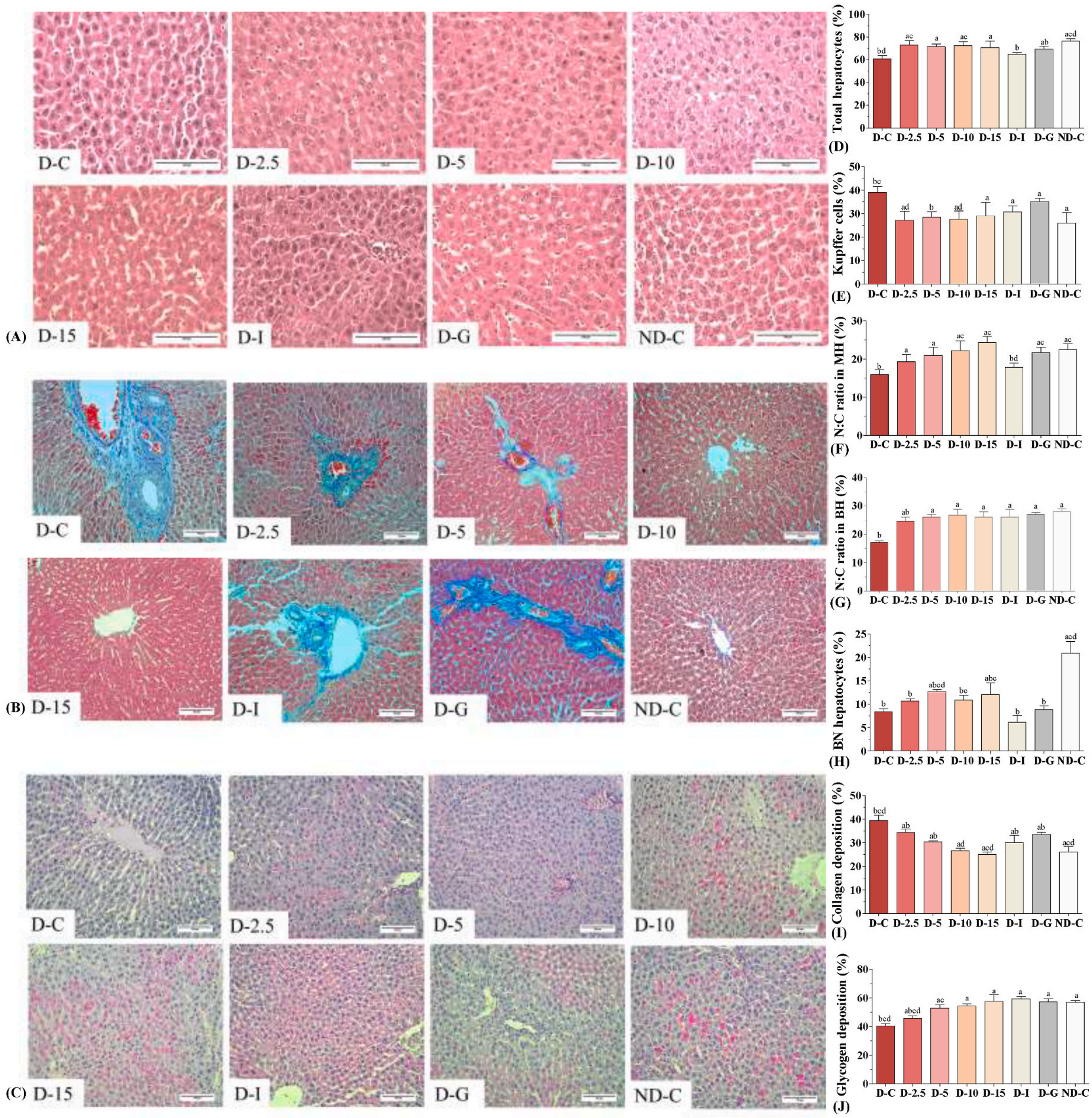

Fig. 11. Histopathological analysis of the H&E (A), Masson's trichrome (B) and PAS (C) stained liver tissue of experimental groups in the anti-diabetic study. Polyherbal mixture effects on the percentage of total hepatocytes (D), the percentage of Kupffer cells (E), nucleus:cytoplasm ratio in mononucleated hepatocytes (F), nucleus:cytoplasm ratio in binucleated hepatocytes (G), percentage of binucleated hepatocytes (H), percentage of collagen deposition (I) and percentage of glycogen deposition (J) in liver. D–C: Diabetic control; D-2.5: Polyherbal mixture 2.5 g/kg; D-5: Polyherbal mixture 5 g/kg; D-10: Polyherbal mixture 10 g/kg; D-15: Polyherbal mixture 15 g/kg; D–I: Insulin; D–G: Glimepiride; ND-C: non-diabetic control. Data were expressed as the mean ± standard deviation (n = 5), [a]$p < 0.001$ compared to the D-C group; [b]$p < 0.001$ compared to the ND-C group; [c]$p < 0.001$ compared to the D-I group; [d]$p < 0.001$ compared to the D-G group.

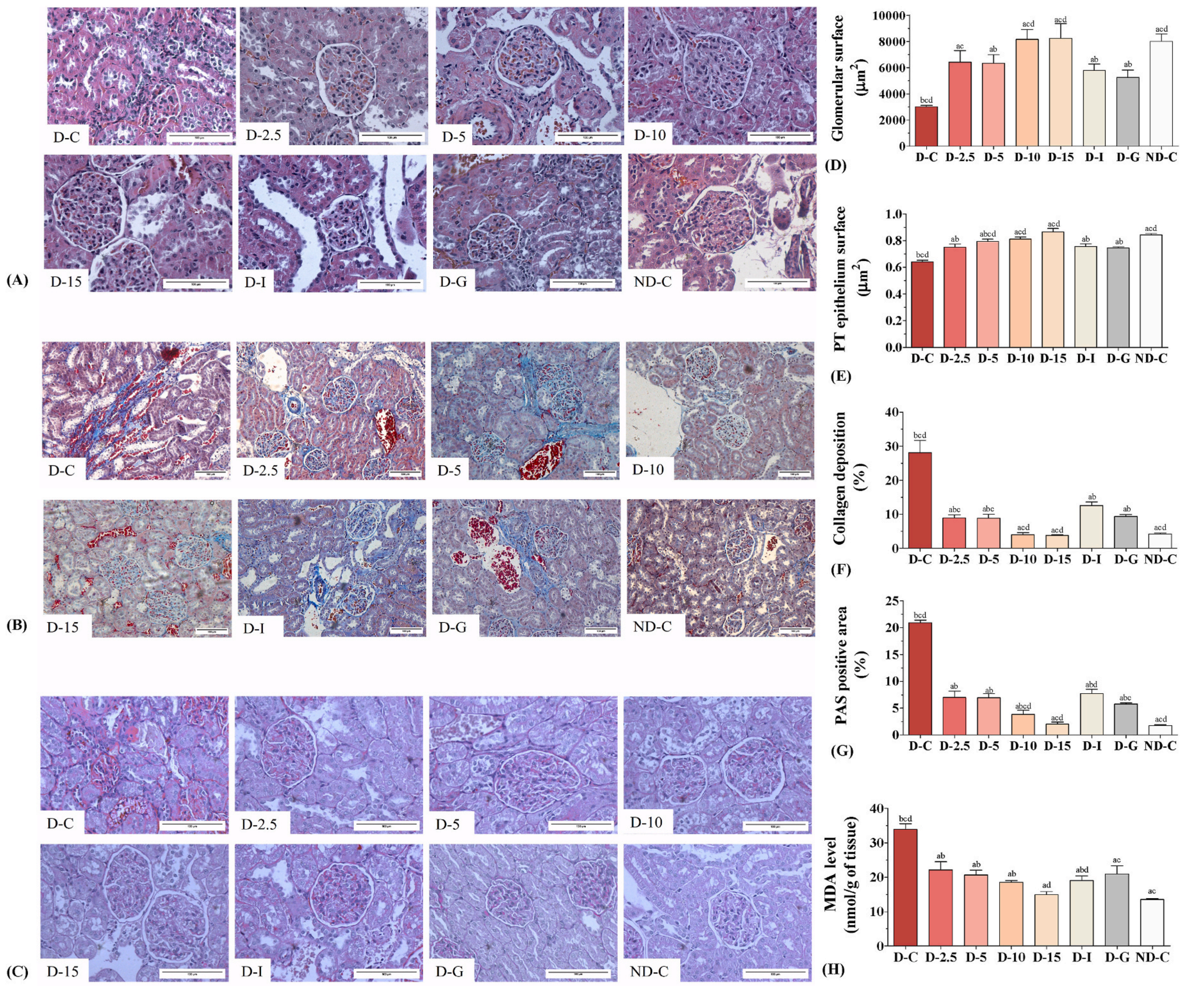

Fig. 12. Histopathological analysis of the H&E (A), Masson's trichrome (B) and PAS (C) stained renal tissue of experimental groups in the anti-diabetic study. Polyherbal mixture effects on the glomerular surface area (D), proximal tubule epithelium surface (E), the percentage of collagen deposition (F), and percentage of PAS positive stained area (G), MDA level (H) in kidney. D–C: Diabetic control; D-2.5: Polyherbal mixture 2.5 g/kg; D-5: Polyherbal mixture 5 g/kg; D-10: Polyherbal mixture 10 g/kg; D-15: Polyherbal mixture 15 g/kg; D–I: Insulin; D–G: Glimepiride; ND-C: non-diabetic control. Data were expressed as the mean ± standard deviation (n = 5), $^{a}p < 0.001$ compared to the D-C group; $^{b}p < 0.001$ compared to the ND-C group; $^{c}p < 0.001$ compared to the D-I group; $^{d}p < 0.001$ compared to the D-G group.

in the liver tissue of the untreated diabetic controls (39.36 ± 2.39%) was significantly increased ($p < 0.001$) compared to the healthy animals (26.06 ± 2.12%) (Fig. 11 (I)), i.e., alloxan-induced diabetes led to the development of hepatic fibrosis (Liu et al., 2010; Popescu et al., 2012; Ozbek et al., 2017; Wilkinson et al., 2019; Madić et al., 2021).

Also, alloxan-induced diabetes also resulted in an imbalance of liver metabolism, which was evident in decreased liver glycogen in diabetic controls (40.25 ± 1.84%) compared to the healthy ones (56.92 ± 1.04%) (Fig. 11 (J)) (Greenberg et al., 2006).

Treatment with the polyherbal mixture significantly reduced liver impairment, primarily in terms of decreasing the deposition of collagen and the number of Kupffer cells, and increasing the number of hepatocytes, especially the percent of binucleated ones (Fig. 11), indicating an increase in the division frequency of hepatocytes, due to the presence of caffeic acid and epicatechin (Quine and Raghu, 2005; Celik et al., 2009). Similar to the study of Aldayel and associates (2020), the same duration of treatment with insulin and glimepiride also managed to reduce the impairment of the liver, but less successfully than treatment with the highest concentration of herbal mixture tested (Fig. 11).

Namely, all tested concentrations of the polyherbal mixture successfully increased the percent of total hepatocytes (72.93 ± 4.07, 71.63 ± 2.23, 72.43 ± 3.58, and 70.90 ± 5.67% in experimental groups D-2.5, D-5, D-10, and D-15, respectively) compared to diabetic controls ($p < 0.001$) (Fig. 11 (D)). At the same time, all tested concentrations of the extract significantly decreased the percent of Kupffer cells (27.06 ± 4.07, 28.47 ± 2.29, 27.58 ± 3.58, and 29.09 ± 5.67% in groups D-2.5, D-5, D-10, and D-15, respectively) compared to the D-C group ($p < 0.001$) (Fig. 11 (E)).

Also, treatment with all tested concentrations of the polyherbal mixture, as well as treatments with insulin and glimepiride, completely normalized the N:C ratio in binucleated hepatocytes (26.11 ± 1.00, 26.78 ± 2.05, 26.14 ± 1.76, 26.07 ± 2.81, and 27.10 ± 0.60% in experimental groups D-5, D-10, D-15, D-I, and D-G, respectively) (Fig. 11 (G)). The highest tested concentration of polyherbal mixture (15 g/kg) managed to increase the percent of binucleated hepatocytes (12.10 ± 2.45%) and to fully restore the N:C ratio in mononuclear hepatocytes to the level of healthy animals (24.31 ± 1.62) ((Fig. 11 (H, F)).

Treatments with 10 and 15 g/kg of herbal mixture completely normalized the deposition of collagen in the liver (26.65 ± 0.99 and 25.06 ± 0.96%, respectively) (Fig. 11 (I)), whereas treatments with the highest tested concentration of the polyherbal mixture (15 g/kg) and treatments with insulin and glimepiride managed to normalize the deposition of glycogen in liver tissue as well (57.43 ± 4.77, 59.23 ± 1.79, and 57.04 ± 2.36% in D-15, D-I, and D-G, respectively) ($p < 0.001$) (Fig. 11 (J)).

Glimepiride treatment increased glycogen deposition (Fig. 11 (J)), through the expression of GLUT4 protein activity (Mori et al., 2008). On the other hand, treatment with the polyherbal mixture increased the deposition of liver glycogen, through the regeneration of β-cells (Supplementary Fig. 6 (C)). Namely, it is well known that hydroxybenzoic acid, one of the bioactive compounds of the tested polyherbal mixture, is capable of regenerating β-cells, and normalize levels of serum insulin and liver glycogen (Peungvich et al., 1998). Besides that, the observed hepatoprotective activity of the tested polyherbal mixture can be explained by its high antioxidant activity that enhanced antioxidative defense enzymes via Nrf2/cytochrome P450 2E1 (CYP2E1) expression, decreased inflammation through inactivation of mitogen-activated protein kinase (MAPK)/NF-κB signalling pathways and reduced apoptosis through regulating B-cell lymphoma 2 (Bcl-2)/protein kinase B (AKT)/caspase expression (Li et al., 2018).

3.3.2.5. The effect of the polyherbal mixture on diabetic animals' kidney histology. Histopathological analysis of renal tissue confirmed that alloxan-induced diabetes resulted in a significant deterioration of renal tissue. Namely, as presented in Fig. 12 (D, E), both the glomerular surface area and the surface area of the epithelium were significantly lower ($p < 0.001$) in the D-C group (3017.45 ± 82.54 and 0.64 ± 0.01 μm^2, respectively) compared to the ND-C group (8010.57 ± 481.24 and 0.84 ± 0.01 μm^2, respectively), suggesting that the osmotic pressure in the blood vessels of the glomerulus led to glomerulosclerosis and disruption of the normal process of blood plasma filtration (Alsaad and Herzenberg, 2007; Baues et al., 2020; Bouteldja et al., 2021), while the decreased surface area of the walls of the proximal tubules (Fig. 12 (E)) indicates damage of the renal tubule epithelium and disruption of reabsorption in the kidneys.

Moreover, collagen deposition and the percent of PAS-positive stained area in the kidneys of the diabetic controls (20.95 ± 0.39 and 28.09 ± 3.12%, respectively) were significantly higher ($p < 0.001$) compared to the healthy controls (1.76 ± 0.11 and 4.17 ± 0.26%, respectively) (Fig. 12 (F, G)), which is additional evidence of significant nephropathy in terms of fibrosis and additional proof that diabetes was well developed in experimental animals (Helen et al., 1994; Kataya et al., 2011).

In addition, the high MDA content in the renal homogenate (Fig. 12 (H)) denotes increased degradation of lipids in the kidney itself, caused by epithelial membrane damages (Oktem et al., 2005; Gowda et al., 2010).

Treatment with higher tested doses of the polyherbal mixture (10 and 15 g/kg) completely reduced alloxan-induced diabetic nephropathy in terms of restoring both glomerular size (8161.34 ± 651.68 and 8249.81 ± 964.82 μm^2, respectively) and collagen deposition (4.01 ± 0.54 and 3.80 ± 0.18% respectively) to the normal levels (Fig. 12 (D, F)). Furthermore, treatment with the highest tested concentration of the herbal mixture (15 g/kg) successfully normalized the epithelial surface area of proximal tubules (0.86 ± 0.02 μm2) (Fig. 12 (E)), indicating regeneration of kidney epithelia (Kataya et al., 2011). Furthermore, the highest tested concentration of the polyherbal mixture decoction (15 g/kg), through normalization of glycemic values (Fig. 7), decreased the level of filtrated sugar visible as PAS-positive areas in kidney tissue (2.01 ± 0.37%) (Fig. 12 (G)) to the level of healthy controls (Fig. 12 (G)). Treatments with insulin and glimepiride also showed some nephroprotective activities, but they did not lead to the complete regeneration of kidney tissue observed in diabetic animals treated with the highest tested concentration of the polyherbal mixture.

Taking into account the results of the compound analysis of the tested polyherbal mixture (Table 2, Fig. 1), the mechanism of this polyherbal mixture decoct is well described in the literature. Namely, caffeic acid decreases the level of MDA in the kidney and visibly reduces renal epithelial damage in animal diabetic model, while administration of rutin regulates renal function and reduces the degree of renal tissue damage in induced nephropathy by down-regulating the expression of TGFβ-1 and fibronectin (Ganesan et al., 2020; Oršolić et al., 2021). The use of epicatechin also reduces lipid peroxidation in the kidneys due to the presence of benzo rings and aromatic compounds that can bind hydroxylated radicals in tissues (Quine and Raghu, 2005). The use of hyperoside in the treatment of diabetic nephropathy regulates blood urea and creatinine levels and reduces renal tissue damage by suppressing expression of fibronectin, collagen IV, tissue inhibitors of metalloproteinase 1 (TIMP-1) expressions while promoting matrix metalloproteinases-9 and -2 (MMP-9 and MMP-2) expressions. (Zhang et al., 2016). It also prevents further apoptosis of podocytes in the glomerulus, which occurs as a result of diabetes, enabling regeneration of damaged tissue via miR-499–5p/APC axis (Zhou et al., 2021). Moreover, isoquercetin possesses its nephroprotective ability through its hypoglycemic, hypolipidemic (Zhang et al., 2011; Jayachandran et al., 2018), and hepatoprotective activities, which are similar to those of sulfonylureas (Jayachandran et al., 2018). In addition, caffeic acid regulates lipid status and blood glucose level and attenuates oxidative damage of blood cells, liver cells, and kidney tissue (Oršolić et al., 2021) by upregulating nuclear erythroid-related factor 2 and downregulating nuclear factor-kappa B (Ajiboye et al., 2019).

3.3.2.6. The effect of the polyherbal mixture on diabetic animals' hippocampal histology. The number of viable neurons in all four hippocampal regions was significantly decreased ($p < 0.001$) in diabetic controls (33.66 ± 5.26, 26.73 ± 4.67, 48.03 ± 3.17 and 18.06 ± 4,83% in hippocampal regions CA1, CA2, CA3, and CA4, respectively) compared to the healthy controls (99.25 ± 1.29, 97.71 ± 2.32, 95.98 ± 4.83, and 96.26 ± 3.47% in the CA1, CA2, CA3, and CA4 regions of the hippocampus, respectively), indicating a possible risk of developing Alzheimer's disease (Shieh et al., 2020). This is in concordance with a number of studies that suggest that memory loss is associated with both type 1 and type 2 diabetes (Biessels and Gispen, 2005; Beauquis et al., 2010). To wit, prolonged hyperglycemia and increased oxidative stress may lead to loss of hippocampal pyramidal cells, changes in synaptic plasticity, difficulties in forming new and long-term memories and, finally, memory loss (Kamal et al., 2003; Biessels and Gispen, 2005; Tomlinson and Gardiner, 2008; Bree et al., 2009; Hori et al., 2010; Rasgon et al., 2011; Zhang et al., 2015).

Of all the treatments, only the highest tested concentration of the polyherbal mixture brought the number of viable neurons in all regions of the hippocampus to the level of healthy animals (96.71 ± 2.78, 97.07 ± 0.69, 95.87 ± 1.66, and 93.17 ± 0.71% in CA1, CA2, CA3, and CA4 hippocampal regions, respectively) (Fig. 13 (E, F, G, H)). Interestingly, glimepiride treatment was able to normalize the number of viable neurons only in the CA1 region (96.16 ± 3.84%) (Fig. 13 (E)), whereas insulin treatment restored this parameter only in the CA2 region (89.37 ± 8.77%) (Fig. 13 (F)), similarly to the study of Osborne and associates (2016).

Considering this and the fact that this concentration of the polyherbal decoction was able to completely normalize the glycemic levels of the diabetic animals (Fig. 7), we might conclude that it can prevent the potential loss of hippocampal neurons by maintaining normoglycemia.

These results agree with previous studies where diets enriched with natural polyphenols ameliorate the loss of neurons and prevent dementia and Alzheimer's disease (Ahmed et al., 2012). In addition to a direct effect on reducing oxidative stress, polyphenols improve learning and memory by regulating signaling pathways in nerve cells. To wit, hyperoside, one of the most abundant polyphenols in this polyherbal mixture decoction, stimulates the activity of the AMPA glutamate receptor and the expression of brain-derived neurotrophic factor (BDNF) in the hippocampus (Nichols et al., 2015; Gong et al., 2017; Chen et al., 2017; Yi et al., 2022). Epicatechin reduces the risk of memory loss by lowering reactive oxygen radical levels in the brain (Cuevas et al., 2009). Moreover, quercetin reduces the accumulation of β-amyloid

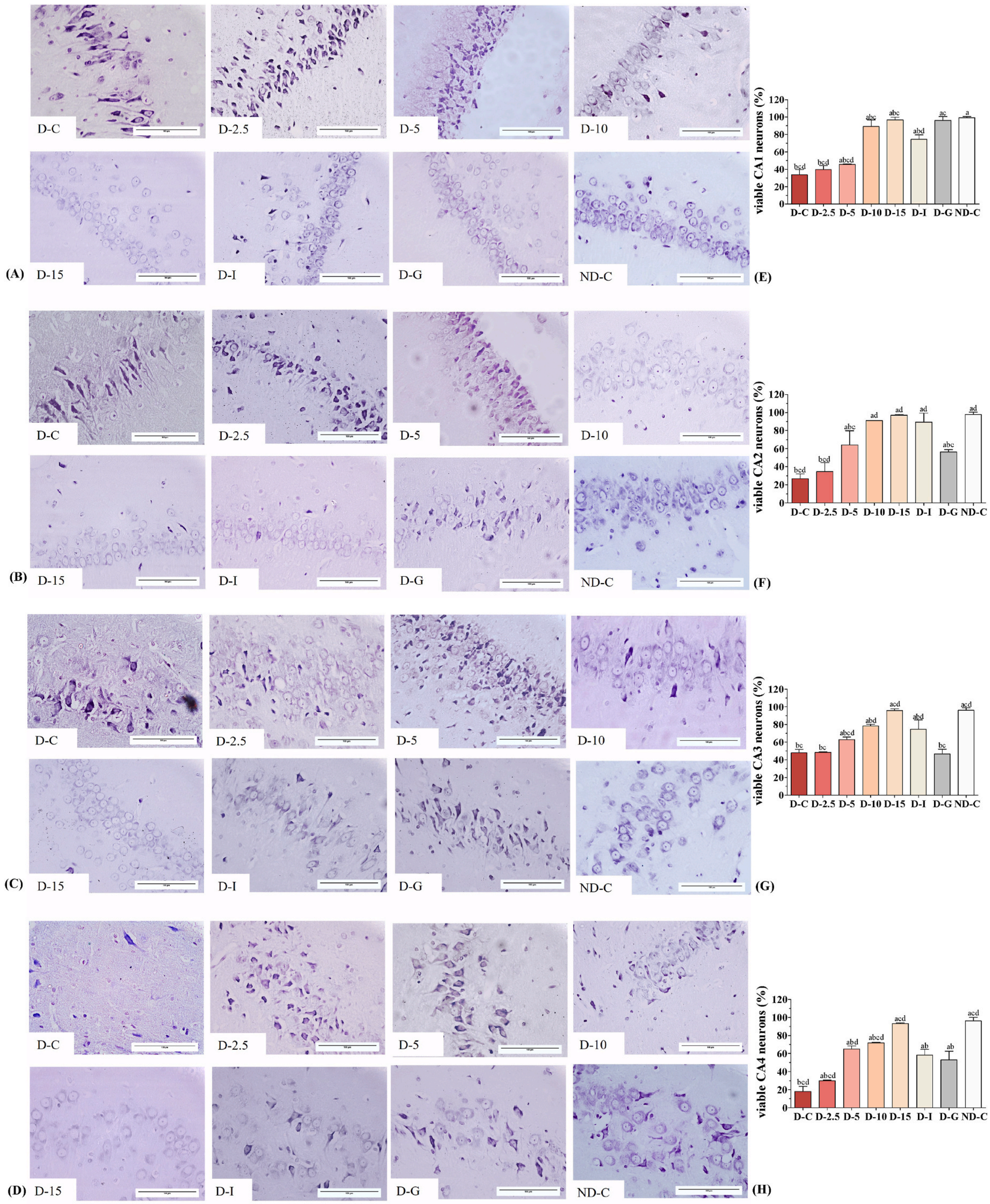

Fig. 13. Histopathological analysis of the Nissl stained CA1 (A), CA2 (B), CA3 (C) and CA4 (D) hippocampal regions of experimental groups in the anti-diabetic study. Polyherbal mixture effects on the percentage of the viable neurons in CA1 (E), CA2 (F), CA3 (G), and CA4 (H) hippocampal regions. D–C: Diabetic control; D-2.5: Polyherbal mixture 2.5 g/kg; D-5: Polyherbal mixture 5 g/kg; D-10: Polyherbal mixture 10 g/kg; D-15: Polyherbal mixture 15 g/kg; D–I: Insulin; D–G: Glimepiride; ND-C: non-diabetic control. Data were expressed as the mean ± standard deviation (n = 5), $^{a}p < 0.05$ compared to the D-C group; $^{b}p < 0.05$ compared to the ND-C group; $^{c}p < 0.05$ compared to the D-I group; $^{d}p < 0.05$ compared to the D-G group.

plaques and inhibits interactions of microglia and astrocytes via downregulation of IL-1β/COX-2/iNOS inflammatory signaling (Vargas-Restrepo et al., 2018). Furthermore, isoquercetin and rutin decrease inflammation through suppression of the cytokines TNF-α and IL-6; via upregulation of Nrf2 expression via ERK1/2-JNK and p38 signaling and by deactivation of TLR4-NF-κB (Wang et al., 2013, 2017; Song et al., 2014; Chen et al., 2017; Dai et al., 2018; Nkpaa et al., 2019; Çelik et al., 2020). In addition, caffeic acid regulates the activation of Nrf2 (Huang et al., 2013; Morroni et al., 2018) and activates the PKA/CREB pathway in neurons (Javed et al., 2012; Fu et al., 2017; Parashar et al., 2017; Koga et al., 2019; Anjomshoa et al., 2020).

To the best of our knowledge, this is the first report on the *in vitro* and *in vivo* evaluation of this polyherbal mixture decoction.

4. Conclusions

The polyherbal mixture used for ethnopharmacological purposes was not toxic at any of the tested concentrations and it was effective in preventing both primary and secondary diabetic complications such as hyperlipidemia, increased lipid peroxidation, non-alcoholic fatty liver disease, nephropathies and neurodegeneration in a diabetic rat model. These results suggest that this mixture might be a valuable candidate for future use as a dietary supplement. However, more comprehensive studies which will further elucidate the exact mechanism of the therapeutic potential of the polyherbal mixture are needed.

Authors' contributions

Aleksandra Petrović;: Literature research. Designing a study. Experimental work (*in vitro* and *in vivo* experiments). Protocols modifications. Microscopy. Histopathology. Analyzing the results. Writing the paper. Design of graphical abstract. Dr. Višnja Madić;: Histopathology. Writing the paper. Dr. Gordana Stojanović;: HPLC-UV analysis. Review of the paper. Dr. Ivana Zlatanović;: HPLC-UV analysis. Review of the paper. Dr. Bojan Zlatković;: Plant collecting. Identification. Review of the paper. Dr. Perica Vasiljević;: Confirmation of *in vitro* experimental results. Review of the paper. Dr. Ljubiša Đorđević;: Supervision of all the work.

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Declaration of competing interest

The authors declare that there is no conflict of interest.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jep.2023.117032.

References

Ahmed, D., Sharma, M., Mukerjee, A., Kamal Kant, R., Kumar, V., 2012. Antidiabetic, anti-hyperlipidemic & hepatoprotective effect of a polyherbal unani formulation “qurs tabasheer” in STZ-diabetic wistar rats introduction. Nat. Preced. https://doi.org/10.1038/npre.2012.7056.1.

Ahmed, N., 2009. Alloxan diabetes-induced oxidative stress and impairment of oxidative defense system in rat brain: neuroprotective effects of *Cichorium intybus*. Int. J. Diabetes Metabol. 17 (3) https://doi.org/10.1159/000497681.

Aldayel, T.S., Alshammari, G.M., Omar, U.M., Grace, M.H., Lila, M.A., Yahya, M.A., 2020. Hypoglycaemic, insulin releasing, and hepatoprotective effect of the aqueous extract of Aloe perryi Baker resin (Socotran Aloe) in streptozotocin-induced diabetic rats. JTUMED 14, 1671–1685. https://doi.org/10.1080/16583655.2020.1855859.

Ali, H., Houghton, P.J., Soumyanath, A., 2006. α-Amylase inhibitory activity of some Malaysian plants used to treat diabetes; with particular reference to *Phyllanthus amarus*. J. Ethnopharmacol. 107, 449–455. https://doi.org/10.1016/j.jep.2006.04.004.

Alsaad, K.O., Herzenberg, A.M., 2007. Distinguishing diabetic nephropathy from other causes of glomerulosclerosis: an update. J. Clin. Pathol. https://doi.org/10.1136/jcp.2005.035592.

Amparo, P.A., Sara, M.E., Markus, P., 2017. Pro- and antioxidant functions of the peroxisome- mitochondria connection and its impact on aging and disease. Oxid. Med. Cell. Longev. 17 https://doi.org/10.1155/2017/9860841, 2017.

Anjomshoa, M., Boroujeni, S.N., Ghasemi, S., Lorigooini, Z., Amiri, A., Balali-Dehkordi, S., Amini-Khoei, H., 2020. Rutin via increase in the CA3 diameter of the Hippocampus exerted antidepressant-like effect in mouse model of maternal separation stress: possible involvement of NMDA receptors. Behav. Neurol. https://doi.org/10.1155/2020/4813616, 2020.

Arvanitakis, Z., Wilson, R., Benias, J.L., Evans, D.A., Bennett, D.A., 2004. Diabetes mellitus and risk of Alzheimer disease and decline in cognitive function. Arc Neurol 61, 661–666. https://doi.org/10.1001/archneur.61.5.661.

Arya, A., Jamil Al-Obaidi, M.M., Shahid, N., Bin Noordin, M.I., Looi, C.Y., Wong, W.F., Khaing, S.L., Mustafa, M.R., 2014. Synergistic effect of quercetin and quinic acid by alleviating structural degeneration in the liver, kidney and pancreas tissues of STZ induced diabetic rats: a mechanistic study. Food Chem. Toxicol. 71, 183–196. https://doi.org/10.1016/j.fct.2014.06.010.

Ajiboye, T.O., Ajala-Lawal, R.A., Adeyiga, A.B., 2019. Caffeic acid abrogates 1, 3-dichloro-2-propanol-induced hepatotoxicity by upregulating nuclear erythroid-related factor 2 and downregulating nuclear factor-kappa B. Hum. Exp. Toxicol. 38 (9), 1092–1101. https://doi.org/10.1177/0960327119851257.

Bastaki, S., 2005. Diabetes mellitus and its treatment. IJDMD 3 (3), 111–134.

Baues, M., Klinkhammer, B.M., Ehling, J., Gremse, F., van Zandvoort, M.A.M.J., Reutelingsperger, C.P.M., Daniel, C., Amann, K., Bábíčková, J., Kiessling, F., Floege, J., Lammers, T., Boor, P., 2020. A collagen-binding protein enables molecular imaging of kidney fibrosis *in vivo*. Kidney Int. 97, 609–614. https://doi.org/10.1016/j.kint.2019.08.029.

Beauquis, J., Roig, P., de Nicola, A.F., Saravia, F., 2010. Short-term environmental enrichment enhances adult neurogenesis, vascular network and dendritic complexity in the hippocampus of type 1 diabetic mice. PLoS One 5 (11), e13993. https://doi.org/10.1371/journal.pone.0013993.

Bhandari, M.R., Jong-Anurakkun, N., Hong, G., Kawabata, J., 2008. α-Glucosidase and α-amylase inhibitory activities of Nepalese medicinal herb Pakhanbhed (*Bergenia ciliate*, Haw.). Food Chem. 106, 247–252. https://doi.org/10.1016/j.foodchem.2007.05.077.

Biessels, G.J., Gispen, W.H., 2005. The impact of diabetes on cognition: what can be learned from rodent models? Neurobiol. Aging 36–41. https://doi.org/10.1016/j.neurobiolaging.2005.08.015.

Blois, M.S., 1958. Antioxidant Determinations by the use of a stable free radical. Nature 181, 1199–1200. https://doi.org/10.1038/1811199a0.

Bouteldja, N., Klinkhammer, B.M., Bülow, R.D., Droste, P., Otten, S.W., von Stillfried, S. F., Moellmann, J., Sheehan, S.M., Korstanje, R., Menzel, S., Bankhead, P., Mietsch, M., Drummer, C., Lehrke, M., Kramann, R., Floege, J., Boor, P., Merhof, D., 2021. Deep learning-based segmentation and quantification in experimental kidney histopathology. JASN (J. Am. Soc. Nephrol.) 32, 52–68. https://doi.org/10.1681/ASN.2020050597.

Bouyahya, A., Belmehdi, O., el Jemli, M., Marmouzi, I., Bourais, I., Abrini, J., Faouzi, M. E.A., Dakka, N., Bakri, Y., 2019. Chemical variability of *Centaurium erythraea* essential oils at three developmental stages and investigation of their in vitro antioxidant, antidiabetic, dermatoprotective and antibacterial activities. Ind. Crops Prod. 132, 111–117. https://doi.org/10.1016/j.indcrop.2019.01.042.

Bree, A.J., Puente, E.C., Daphna-Iken, D., Fisher, S.J., 2009. Diabetes Increases Brain Damage Caused by Severe Hypoglycemia, vol. 297, pp. 194–201. https://doi.org/10.1152/ajpendo.91041.2008, 1.

Brown, A.C., 2017. Kidney toxicity related to herbs and dietary supplements: online table of case reports. Part 3 of 5 series. Food Chem. Toxicol. 107, 502–519. https://doi.org/10.1016/j.fct.2016.07.024. PMID: 28755953.

Cao, Y., Han, S., Lu, H., Luo, Y., Guo, T., Wu, Q., Luo, F., 2022. Targeting mTOR signaling by dietary polyphenols in obesity prevention. Nutrients 14 (23), 5171. https://doi.org/10.3390/nu14235171.

Çelik, H., Kandemir, F.M., Caglayan, C., Özdemir, S., Çomaklı, S., Kucukler, S., Yardım, A., 2020. Neuroprotective effect of rutin against colistin-induced oxidative stress, inflammation and apoptosis in rat brain associated with the CREB/BDNF expressions. Mol. Biol. Rep. 47, 2023–2034. https://doi.org/10.1007/s11033-020-05302-z.

Celik, S., Erdogan, S., Tuzcu, M., 2009. Caffeic acid phenethyl ester (CAPE) exhibits significant potential as an antidiabetic and liver-protective agent in streptozotocin-induced diabetic rats. Pharmacol. Res. 60, 270–276. https://doi.org/10.1016/j.phrs.2009.03.017.

Chang, J.M., Kuo, M.C., Kuo, H.T., Chiu, Y.W., Chen, H.C., 2005. Increased glomerular and extracellular malondialdehyde levels in patients and rats with diabetic nephropathy. J. Lab. Clin. Med146, 210–215. https://doi.org/10.1016/j.lab.2005.05.007.

Chen, D., Chen, X., He, C., Chen, Z., Chen, Q., Chen, J., Bo, H., 2023. Sanhuang xiexin decoction synergizes insulin/PI3K-Akt/FoxO signaling pathway to inhibit hepatic glucose production and alleviate T2DM. J. Ethnopharmacol. 116162 https://doi.org/10.1016/j.jep.2023.116162.

Chen, M., Dai, L.H., Fei, A., Pan, S.M., Wang, H.R., 2017. Isoquercetin activates the ERK1/2-Nrf2 pathway and protects against cerebral ischemia-reperfusion injury in vivo and in vitro. Exp. Ther. Med. 13, 1353–1359. https://doi.org/10.3892/etm.2017.4093.

Cuevas, E., Limón, D., Pérez-Severiano, F., Díaz, A., Ortega, L., Zenteno, E., Guevara, J., 2009. Antioxidant effects of Epicatechin on the hippocampal toxicity caused by Amyloid-beta 25-35 in rats. Eur. J. Pharmacol. 616, 122–127. https://doi.org/10.1016/j.ejphar.2009.06.013.

Dai, Y., Zhang, H., Zhang, J., Yan, M., 2018. Isoquercetin attenuates oxidative stress and neuronal apoptosis after ischemia/reperfusion injury via Nrf2-mediated inhibition of the NOX4/ROS/NF-κB pathway. Chem. Biol. Interact. 284, 32–40. https://doi.org/10.1016/j.cbi.2018.02.017.

Dalar, A., Konczak, I., 2014. Cichorium intybus from Eastern Anatolia: phenolic composition, antioxidant and enzyme inhibitory activities. Ind. Crops Prod. 60, 79–85. https://doi.org/10.1016/j.indcrop.2014.05.043.

Daraz, S.N., Abo-zid, M.M., Ally, A.F., El-Debis, A.A., 2010. Hypoglycemic and hypolipidemic effect of chicory (*Cichorium intybus* L.) herb in diabetic rats. Minufiya J. Agric. Res. 35 (4), 1201–1208.

Demir, E., Ozkan, H., Seckin, K.D., Sahtiyanc, B., Demir, B., Tabak, O., Kumbasar, A., Uzun, H., 2019. Plasma Yonulin levels as a Non-Invasive biomarker of intestinal permeability in women with gestational Diabetes Mellitus. Biomoleculs 9 (1). https://doi.org/10.3390/biom9010024.

Đorđević, M., Grdović, N., Mihailović, M., Jovanović, J.A., Uskoković, A., Rajić, J., Dordević, Marija, Tolić, A., Mišić, D., Šiler, B., Poznanović, G., Vidaković, M., Dinić, S., 2020. *Centaurium erythraea* extract reduces redox imbalance and improves insulin expression and secretion in pancreatic β-cells exposed to oxidative and nitrosative stress. Arch. Biol. Sci. 72, 117–128. https://doi.org/10.2298/ABS200127005D.

Dowarah, J., Singh, V.P., 2020. Anti-diabetic drugs approaches and advancements. Bioorg. Med. Chem. 28 (5), 115263 https://doi.org/10.1016/j.bmc.2019.115263.

Elangovan, A., Subramanian, A., Durairaj, S., Ramachandran, J., Lakshmanan, D.K., Ravichandran, G., Nambirajan, G., Thilagar, S., 2019. Antidiabetic and hypolipidemic efficacy of skin and seed extracts of Momordica cymbalaria on alloxan induced diabetic model in rats. J. Ethnopharmacol. 241 https://doi.org/10.1016/j.jep.2019.111989.

Evans, J.L., Goldfine, I.D., Maddux, B.A., Grodsky, G.M., 2002. Oxidative stress and stress-activated signaling pathways: a unifying hypothesis of type 2 diabetes. Endocr. Rev. 23, 599–622.

Fang, J.-Y., Zhu, L., Yi, T., Zhang, J.-Y., Yi, L., Liang, Z.-T., Xia, L., Feng, J.-F., Xu, H., Tang, Y.-N., Zhao, Z.-Z., Chen, H.-B., 2015. Fingerprint analysis of processed Rhizoma Chuanxiong by high-performance liquid chromatography coupled with diode array detection. Chin. Med. 10 (1), 1–7.

Feng, J., Ren, H., Gou, Q., Zhu, L., Ji, H., Yi, T., 2016. Comparative analysis of the major constituents in three related polygonaceous medicinal plants using pressurized liquid extraction and HPLC-ESI/MS. Anal. Methods 8 (7), 1557–1564.

Friedewald, W.T., Levy, R.I., Fredrickson, D.S., 1972. Determination of LDL cholesterol. In: Gaw, A. (Ed.), Textbook of Clinical Biochemistry. Churchill Livingstone, London.

Fu, W., Wang, H., Ren, X., Yu, H., Lei, Y., Chen, Q., 2017. Neuroprotective effect of three caffeic acid derivatives via ameliorate oxidative stress and enhance PKA/CREB signaling pathway. Behav. Brain Res. 328, 81–86. https://doi.org/10.1016/j.bbr.2017.04.012.

Ganesan, D., Albert, A., Paul, E., Ananthapadmanabhan, K., Andiappan, R., Sadasivam, S. G., 2020. Rutin ameliorates metabolic acidosis and fibrosis in alloxan induced diabetic nephropathy and cardiomyopathy in experimental rats. Mol. Cell. Biochem. 471, 41–50. https://doi.org/10.1007/s11010-020-03758-y.

Ganesan, K., Xu, B., 2019. Anti-diabetic effects and mechanisms of dietary polysaccharides. Molecules 24 (14), 2556. https://doi.org/10.3390/molecules24142556.

Gholami, H., Saharkhiz, M.J., Fard, F.R., Ghani, A., Nadaf, F., 2018. Humic acid and vermicompost increased bioactive components, antioxidant activity and herb yield of Chicory (*Cichorium intybus* L.). ISBAB 14, 286–292.

Gong, Y., Yang, Y., Chen, X., Yang, M., Huang, D., Yang, R., Zhou, L., Li, C., Xiong, Q., Xiong, Z., 2017. Hyperoside protects against chronic mild stress-induced learning and memory deficits. Biomed. Pharmacother. 91, 831–840. https://doi.org/10.1016/j.biopha.2017.05.019.

Gospodinova, Z., Krasteva, M., 2015. *Cichorium intybus* L. from Bulgaria inhibits viability of human breast cancer cells *in vitro*. Genet. Plant Physiol. 5 (1), 15–22. http://www.bio21.bas.bg/ippg/bg/wp-co.

Gowda, S., Desai, P.B., Kulkarni, S.S., Hull, V.V., Math, A.A.K., Vernekar, S.N., 2010. Markers of renal function tests. N Am. J. Med. 2 (4), 170–173.

Greenberg, C.C., Jurczak, M.J., Danos, A.M., Brady, M.J., 2006. Glycogen branches out: new perspectives on the role of glycogen metabolism in the integration of metabolic pathways glycogen structure and key enzymes of its metabolism. Am. J. Physiol. Endocrinol. Metab. 291, 1–8. https://doi.org/10.1152/ajpendo.00652.2005.-Glycogen.

Gupta, A., Kumar, R., Pandey, A.K., 2020. Antioxidant and antidiabetic activities of *Terminalia bellirica* fruit in alloxan induced diabetic rats. South Afr. J. Bot. 130, 308–315. https://doi.org/10.1016/j.sajb.2019.12.010.

Gurzov, E.N., Eizirik, D.L., 2011. Bcl-2 proteins in diabetes: mitochondrial pathways of β-cell death and dysfunction. Trends Cell Biol. https://doi.org/10.1016/j.tcb.2011.03.001.

Guven, A., Yavuz, O., Cam, M., Ercan, F., Bukan, N., Comunoglu, C., Gokce, F., 2006. Effects of melatonin on streptozotocin-induced diabetic liver injury in rats. Acta Histochem. 108, 85–93. https://doi.org/10.1016/j.acthis.2006.03.005.

Helen, V., Richard, B., Lihare, S., 1994. Biology of disease: pathogenic effect of advanced glycosylation: biochemical, biologic and clinical implications for diabetes and aging. Lab. Invest. 70, 138–148.

Herawati, E.R.N., Santosa, U., Sentana, S., Ariani, D., 2020. Protective Effects of Anthocyanin extract from Purple Sweet potato (*Ipomoea batatus* L.) on blood MDA level, liver and renal activity, and blood pressure of hyperglycemic rats. Prev. Nutr. Food Sci. 25 (4), 375–379. https://doi.org/10.3746/pnf.2020.25.4.375.

Hori, N., Kadota, M.T., Watanabe, M., Ito, Y., Akaike, N., Carpenter, D.O., 2010. Neurotoxic effects of methamphetamine on rat hippocampus pyramidal neurons. Cell. Mol. Neurobiol. 30, 849–856. https://doi.org/10.1007/s10571-010-9512-1.

Huang, R., Wang, B., He, J., Zhang, Z., Xie, R., Li, S., Li, Q., Tian, C., Tou, Y., Chen, W., Xiang, M., 2022. Lian-Qu formula treats metabolic syndrome via reducing fat synthesis, insulin resistance and inflammation. J. Ethnopharmacol. 306, 116060 https://doi.org/10.1016/j.jep.2022.116060.

Huang, Y., Jin, M., Pi, R., Zhang, J., Chen, M., Ouyang, Y., Liu, A., Chao, X., Liu, P., Liu, J., Ramassamy, C., Qin, J., 2013. Protective effects of caffeic acid and caffeic acid phenethyl ester against acrolein-induced neurotoxicity in HT22 mouse hippocampal cells. Neurosci. Lett. 535, 146–151. https://doi.org/10.1016/j.neulet.2012.12.051.

International Diabetes Federation, 2021. IDF Diabetes Atlas, tenth ed. International Diabetes Federation, Brussels, Belgium.

Inzucchi, S.E., Bergenstal, R.M., Buse, J.B., Diamant, M., Ferrannini, E., Neuck, M., Peters, A.L., Tsapas, A., Wender, R., Matthews, D.R., 2015. Management of hyperglycemia in type 2 diabetes, 2015: a patient-centered approach: update to a position statement of the American Diabetes as sociation and the European Association for the Study of Diabetes. Diabetes Care 38, 140–149. https://doi.org/10.2337/dc14-2441.

Islam, B., Sharma, C., Adem, A., Aburawi, E., Ojha, S., 2015. Insight into the mechanism of polyphenols on the activity of HMGR by molecular docking. Drug Des. Dev. Ther. 9, 4943–4951. https://doi.org/10.2147/DDDT.S86705.

Ito, F., Sono, Y., Ito, T., 2019. Measurement and clinical significance of lipid peroxidation as a biomarker of oxidative stress: oxidative stress in diabetes, atherosclerosis, and chronic inflammation. Antioxidants 8. https://doi.org/10.3390/antiox8030072.

Jadhav, R., Puchchakayala, G., 2012. Hypoglycemic and antidiabetic activity of flavonoids: boswellic acid, ellagic acid, quercetin, rutin on streptozotocin-nicotinamide induced type 2 diabetic rats. Int. J. Pharm. Pharmaceut. Sci. 4 (2), 251–256.

Javed, H., Khan, M.M., Ahmad, A., Vaibhav, K., Ahmad, M.E., Khan, A., Ashafaq, M., Islam, F., Siddiqui, M.S., Safhi, M.M., Islam, F., 2012. Rutin prevents cognitive impairments by ameliorating oxidative stress and neuroinflammation in rat model of sporadic dementia of Alzheimer type. Neuroscience 210, 340–352. https://doi.org/10.1016/j.neuroscience.2012.02.046.

Jayachandran, M., Zhang, T., Ganesan, K., Xu, B., Chung, S.S.M., 2018. Isoquercetin ameliorates hyperglycemia and regulates key enzymes of glucose metabolism via insulin signaling pathway in streptozotocin-induced diabetic rats. Eur. J. Pharmacol. 829, 112–120. https://doi.org/10.1016/j.ejphar.2018.04.015.

Kachmar, M.R., Oliveira, A.P., Valentão, P., GilIzquierdoc, A., Domínguez-Perlesc, R., Ouahbia, A., ElBadaouia, K., Andradeb, P.B., Ferreresc, F., 2019. HPLC-DAD-ESI/MSn phenolic profile and in vitro biological potential of *Centaurium erythraea* Rafn aqueous extract. Food Chem. 278, 424–433.

Kamal, A., Artola, A., Biessels, G.J., Gispen, W.H., Ramakers, G.M.J., 2003. Increased spike broadening and slow afterhyperpolarization in CA1 pyramidal cells of streptozotocin-induced diabetic rats. Neuroscience 118, 577–583. https://doi.org/10.1016/S0306-4522(02)00874-6.

Karsdal, M.A., Daniels, S.J., Holm Nielsen, S., Bager, C., Rasmussen, D.G.K., Loomba, R., Surabattula, R., Villesen, I.F., Luo, Y., Shevell, D., Gudmann, N.S., Nielsen, M.J., George, J., Christian, R., Leeming, D.J., Schuppan, D., 2020. Collagen biology and non-invasive biomarkers of liver fibrosis. Liver Int. 40, 736–750. https://doi.org/10.1111/liv.14390.

Kataya, H.H., Hamza, A.E.A., Ramadan, G.A., Khasawneh, M.A., 2011. Effect of licorice extract on the complications of diabetes nephropathy in rats. Drug Chem. Toxicol. 34 (2), 101–108. https://doi.org/10.3109/01480545.2010.510524.

Kim, M.J., Ha, B.J., 2013. Antihyperglycemic and antihyperlipidemic effects of fermented *Rhynchosia nulubilis* in alloxan-induced diabetic rats. Toxicol. Res. 29, 15–19. https://doi.org/10.5487/TR.2013.29.1.015.

Kirpichnikov, D., Samy, Mcfarlane, I., Sowers, J.R., 2002. Metformin: an update. Ann. Intern. Med. 137 (1), 25–33. https://doi.org/10.7326/0003-4819-137-1-200207020-00009.

Kobayashi, S., 2019. The effect of polyphenols on hypercholesterolemia through inhibiting the transport and expression of Niemann–pick C1-like 1. Int. J. Mol. Sci. https://doi.org/10.3390/ijms20194939.

Koga, M., Nakagawa, S., Kato, A., Kusumi, I., 2019. Caffeic acid reduces oxidative stress and microglial activation in the mouse hippocampus. Tissue Cell 60, 14–20. https://doi.org/10.1016/j.tice.2019.07.006.

Latha, R.C.R., Daisy, P., 2011. Insulin-secretagogue, antihyperlipidemic and other protective effects of gallic acid isolated from Terminalia bellerica Roxb. In streptozotocin-induced diabetic rats. Chem. Biol. Interact. 189, 112–118. https://doi.org/10.1016/j.cbi.2010.11.005.

Lee, Y.E., Yoo, S.H., Chung, J.O., Park, M.Y., Hong, Y.D., Park, S.H., Park, T.S., Shim, S. M., 2020. Hypoglycemic effect of soluble polysaccharide and catechins from green tea on inhibiting intestinal transport of glucose. J. Sci. Food Agric. 100, 3979–3986. https://doi.org/10.1002/jsfa.10442.

Li, S., Tan, H.Y., Wang, N., Cheung, F., Hong, M., Feng, Y., 2018. The potential and action mechanism of polyphenols in the treatment of liver diseases. Oxid. Med. Cell. Longev. https://doi.org/10.1155/2018/8394818, 2018.

Li, F., Feng, K.-L., Zang, J.-C., He, Y.-S., Guo, H., Wang, S.-P., Gan, E.-Y., 2021. Polysaccharides from dandelion (*Taraxacum magnolicum*) leaves: insights into innovative drying techniques on their structural characterizations and biological activities. Int. J.Bio Macromolecules 167, 995–1005. https://doi.org/10.1016/j.ijbiomac.2020.11.054.

Liu, Cheng, Tao, Q., Sun, M., Wu, J.Z., Yang, W., Jian, P., Peng, J., Hu, Y., Liu, Chenghai, Liu, P., 2010. Kupffer cells are associated with apoptosis, inflammation and fibrotic effects in hepatic fibrosis in rats. Lab. Invest. 90, 1805–1816. https://doi.org/10.1038/labinvest.2010.123.

Liu, H., Wang, Q., Liu, Y., Chen, G., Cui, J., 2013. Antimicrobial and antioxidant activities of *cichorium intybus* root extract using orthogonal matrix design. J. Food Sci. 78, 2. https://doi.org/10.1111/1750-3841.12040.

Lucchesi, A.N., Tavares De Freitas, N., Langoni, L., Iii, C., Fernando, S., Marques, G., Spadella, T., 2013. Diabetes mellitus triggers oxidative stress in the liver of alloxan-treated rats: a mechanism for diabetic chronic liver disease 1. Acta Cir. Bras. https://doi.org/10.1590/s0102-86502013000700005.

Madić, V., Petrović, A., Jušković, M., Jugović, D., Djordjević, L., Stojanović, G., Vasiljević, P., 2021. Polyherbal mixture ameliorates hyperglycemia, hyperlipidemia and histopathological changes of pancreas, kidney and liver in a rat model of type 1 diabetes. J. Ethnopharmacol. 265 https://doi.org/10.1016/j.jep.2020.113210.

Madić, V., Popović, A.Ž., Vukelić-Nikolić, M., Đorđević, L., Vasiljević, P., 2018. Ethnopharmacological therapies in the treatment of diabetes in Serbia. Glasnik Antropološkog društva Srbije 53, 1–2. https://doi.org/10.5937/gads53-18083.
Madić, V., Stojanović-Radić, Z., Jušković, M., Jugović, D., Žabar Popović, A., Vasiljević, P., 2019. Genotoxic and antigenotoxic potential of herbal mixture and five medicinal plants used in ethnopharmacology. South Afr. J. Bot. 125, 290–297. https://doi.org/10.1016/j.sajb.2019.07.043.
Mahajan, N.M., Lokhande, B.B., Thenge, R.R., Gangane, P.S., Dumore, N.G., 2018. Polyherbal formulation containing antioxidants may serve as a prophylactic measure to diabetic cataract: preclinical investigations in rat model. Phcog. Mag. 14 (58), 572. https://doi.org/10.4103/pm.pm_622_17.
Manna, P., Das, J., Ghosh, J., Sil, P.C., 2010. Contribution of type 1diabetes to rat liver dysfunction and cellular damage via activation of NOS, PARP, IkappaBalpha/NF-kappaB, MAPKs, and mitochondria-dependent pathways: prophylactic roleof arjunolic acid. Free Radic. Biol. Med. 48, 1465–1484. https://doi.org/10.1016/j.freeradbiomed.2010.02.025.
Mapeka, T.M., Sandasi, M., Viljoen, A.M., van Vuuren, S.F., 2022. Optimization of antioxidant synergy in a polyherbal combination by experimental design. Molecules 27 (13), 4196. https://doi.org/10.3390/molecules27134196.
Mechchate, H., Es-safi, I., Haddad, H., Bekkari, H., Grafov, A., Bousta, D., 2021. Combination of Catechin, Epicatechin, and Rutin: optimization of a novel complete antidiabetic formulation using a mixture design approach. J. Nutr. Biochem. 88, 108520 https://doi.org/10.1016/j.jnutbio.2020.108520.
Menyiy, N.E., Guaouguaou, F.-E., Baaboua, A.E., Omari, N.E., Taha, D., Salhi, N., Shariati, M.A., Aanniz, T., Benali, T., Zengin, G., El-Zhazly, M., Chamkhi, I., Bouyahya, A., 2021. Phytochemical properties, biological activities and medicinal use of *Centaurium erythraea* Rafn antidiabetic, dermatoprotective and antibacterial activities. J. Ethnophatmacol. 276, 114171 https://doi.org/10.1016/j.jep.2021.114171.
Middha, S.K., Usha, T., RaviKiran, T., 2012. Influence of *Punica granatum* L. on region specific responses in rat brain during Alloxan-induced diabetes. Asian Pac. J. Trop. Biomed. 2 (2), 905–909. https://doi.org/10.1016/S2221-1691(12)60334-7.
Mihaylova, D., Vrancheva, R., Popova, A., 2019. Phytochemical profile and *in vitro* antioxidant activity of *Centaurium erythraea* Rafn. Bulg. Chem. Commun. 51, 95–100.
Mori, R.C.T., Hirabara, S.M., Hirata, A.E., Okamoto, M.M., Machado, U.F., 2008. Glimepiride as insulin sensitizer: increased liver and muscle responses to insulin. Diabetes Obes. Metabol. 10, 596–600. https://doi.org/10.1111/j.1463-1326.2008.00870.x.
Morroni, F., Sita, G., Graziosi, A., Turrini, E., Fimognari, C., Tarozzi, A., Hrelia, P., 2018. Neuroprotective effect of caffeic acid phenethyl ester in a mouse model of Alzheimer's disease involves Nrf2/HO-1 pathway. Aging Dis. 9, 605–622. https://doi.org/10.14336/AD.2017.0903.
Nair, A.B., Jacob, S., 2016. A simple practice guide for dose conversion between animals and human. J. Basic Clin. Pharm. 7 (2), 27. https://doi.org/10.4103/0976-0105.177703.
Nichols, M., Zhang, J., Polster, B.M., Elustondo, P.A., Thirumaran, A., Pavlov, E.v., Robertson, G.S., 2015. Synergistic neuroprotection by epicatechin and quercetin: activation of convergent mitochondrial signaling pathways. Neuroscience 308, 75–94. https://doi.org/10.1016/j.neuroscience.2015.09.012.
Nkpaa, K.W., Onyeso, G.I., Kponee, K.Z., 2019. Rutin abrogates manganese—induced striatal and hippocampal toxicity via inhibition of iron depletion, oxidative stress, inflammation and suppressing the NF-κB signaling pathway. J. Trace Elem. Med. Biol. 53, 8–15. https://doi.org/10.1016/j.jtemb.2019.01.014.
OECD, 2018. Repeated dose 28-day oral toxicity study in rodents (OECD TG 407). In: Revised Guidance Document 150 on Standardized Test Guidelines for Evaluating Chemicals for Endocrine Disruption. OECD Publishing, Paris. https://doi.org/10.1787/9789264304741-22-en.
Ohkawa, H., Ohishi, N., Yagi, K., 1979. Assay for lipid peroxides in animal tissues by thiobarbituric acid reaction. Anal. Biochem. 95 (2), 351–358. https://doi.org/10.1016/0003-2697(79)90738-3.
Oktem, F., Ozguner, F., Mollaoglu, H., Koyu, A., Uz, E., 2005. Oxidative damage in the kidney induced by 900-MHz-emitted mobile phone: protection by melatonin. Arch. Med. Res. 36, 350–355. https://doi.org/10.1016/j.arcmed.2005.03.021.
Omotoso, O.G., 2018. *Moringa oleifera* ameliorates histomorphological changes associated with cuprizone neurotoxicity in the hippocampal *Cornu ammonis* (CA) 3 region. J. Physiol. Sci. 33 (1), 95–99.
Oršolić, N., Sirovina, D., Odeh, D., Gajski, G., Balta, V., Šver, L., Jembrek, M.J., 2021. Efficacy of Caffeic acid on diabetes and its complications in the mouse. Molecules 26. https://doi.org/10.3390/molecules26113262.
Osborne, C., West, E., Nolan, W., McHale-Owen, H., Williams, A., Bate, C., 2016. Glimepiride protects neurons against amyloid-β-induced synapse damage. Neuropharmacology 101, 225–236. https://doi.org/10.1016/j.neuropharm.2015.09.030.
Ott, A., Stolk, R.P., van Harskamp, F., Pols, H.A.P., Hofman, A., Breteler, M.M.B., 1999. Diabetes mellitus and the risk of dementia: the Rotterdam Study. AAN 53, 1937–1942. https://doi.org/10.1212/wnl.53.9.1937.
Ozbek, H., Acikara, O.B., Keskin, I., Kirmizi, N.I., Ozbilgin, S., Oz, B.E., Kurtul, E., Ozrenk, B.C., Tekin, M., Saltan, G., 2017. Evaluation of hepatoprotective and antidiabetic activity of *Alchemilla mollis*. Biomed. Pharmacother. 86, 172–176. https://doi.org/10.1016/j.biopha.2016.12.005.
Ozer, J., Ratner, M., Shaw, M., Bailey, W., Schomaker, S., 2008. The current state of serum biomarkers of hepatotoxicity. Toxicology 245, 194–205. https://doi.org/10.1016/j.tox.2007.11.021.
Palsamy, P., Sivakumar, S., Subramanian, S., 2010. Resveratrol attenuates hyperglycemia-mediated oxidative stress, proinflammatory cytokines and protects hepatocytes ultrastructure in streptozotocin-nicotinamide-induced experimental diabetic rats. Chem. Biol. Interact. 186, 200–210. https://doi.org/10.1016/j.cbi.2010.03.028.
Pan, S.Y., Zhou, S.F., Gao, S.I., Yu, Z.L., Zhang, S.F., Tang, M.K., Sun, J.N., Ma, D.L., Han, Y.F., Fong, W.F., Ko, K.M., 2016. New perspectives on how to discover drugs from herbal medicines: CAM's outstanding contribution to modern therapeutics. Evid Based Complement Alternat Med. 25, 627375 https://doi.org/10.1155/2013/627375.
Panten, U., Schwanstecher, M., Schwanstecher, C., 1996. Sulfonylurea receptors and mechanism of sulfonylurea action. Exp. Clin. Endocrinol. Diabetes 104 (1), 1–9. https://doi.org/10.1055/s-0029-1211414.
Papatheodorou, K., Banach, M., Bekiari, E., Rizzo, M., Edmonds, M., 2018. Complications of diabetes 2017. J. Diabetes Res. 3086167 https://doi.org/10.1155/2018/3086167.
Parashar, A., Mehta, V., Udayabanu, M., 2017. Rutin alleviates chronic unpredictable stress-induced behavioral alterations and hippocampal damage in mice. Neurosci. Lett. 656, 65–71. https://doi.org/10.1016/j.neulet.2017.04.058.
Parvin, M., Zahra, B., Fereidoun, A., 2014. Functional foods-based diet as a novel dietary approach for management of type 2 diabetes and its complications. World J. Diabetes 5 (3), 267–281.
Petrović, A., Madić, V., Jušković, M., Đorđević, Lj, Vasiljević, P., 2021. Osteoprotective effects of 'anti-diabetic' polyherbal mixture in type 1 diabetic rats. Acta Vet. 71, 256–272. https://doi.org/10.2478/acve-2021-0023.
Peungvich, P., Temsiririrkkul, R., Kurmar Prasain, J., Tezuka, Y., Kadota, S., Thirawarapan, S.S., Watanabe, H., 1998. 4-Hydroxybenzoic acid: a hypoglycemic constituent of aqueous extract of *Pandanus odorus* root. J. Ethnopharmacol. 62 (1), 79–84. https://doi.org/10.1016/s0378-8741(98)00061-0.
Pitchai, D., Manikkam, R., 2012. Hypoglycemic and insulin mimetic impact of catechin isolated from Cassia fistula: a substantiate *in silico* approach through docking analysis. Med. Chem. Res. 21, 2238–2250. https://doi.org/10.1007/s00044-011-9722-1.
Popescu, R., Filimon, M.N., Dumitrescu, G., Ciochina, L.P., Dumitrascu, V., Vlad, D., Verdes, D., 2012. Histological and morphometrical studies in liver regeneration in mice. Sci. Pap.: Animal Sci.Biotechnol. 45 (2), 203–207.
Quine, S.D., Raghu, P.S., 2005. Effects of (-)-epicatechin, a flavonoid on lipid peroxidation and antioxidants in streptozotocin-induces diabetic liver, kidney and heart. Pharmacol. Rep. 54 (5), 610–615.
Rasgon, N.L., Kenna, H.A., Wroolie, T.E., Kelley, R., Silverman, D., Brooks, J., Williams, K.E., Powers, B.N., Hallmayer, J., Reiss, A., 2011. Insulin resistance and hippocampal volume in women at risk for Alzheimer's disease. Neurobiol. Aging 32, 1942–1948. https://doi.org/10.1016/j.neurobiolaging.2009.12.005.
Rice-Evans, C., Omorphos, S.C., Baysal, E., 1986. Sickle cell membranes and oxidative damage. Biochem. J. 237 (1), 265–269. https://doi.org/10.1042/bj2370265.
Rohilla, A., Ali, S., 2012. Alloxan induced diabetes: mechanisms and effects 3. Int. J. Res. Pharm. Biomed. Sci. 3 (2), 2229–3701.
Russell-Jones, D., Khan, R., 2007. Insulin-associated weight gain in diabetes – causes, effects and coping strategies. Diabetes Obes. Metabol. 9, 799–812. https://doi.org/10.1111/j.1463-1326.2006.00686.x.
Savage, D., Petersen, K.F., Shulman, G., 2007. Disordered lipid metabolism and the pathogenesis of insulin resistance. Am. Physiol. Soc. 87, 507–520. https://doi.org/10.1152/physrev.00024.2006.
Schaalan, M., El-Abhar, H.S., Barakat, M., El-Denshary, E.S., 2009. Westernized-like-diet-fed rats: effect on glucose homeostasis, lipid profile, and adipocyte hormones and their modulation by rosiglitazone and glimepiride. J. Diabet. Complicat. 23, 199–208. https://doi.org/10.1016/j.jdiacomp.2008.02.003.
Schloot, N.C., Haupt, A., Schütt, M., Badenhoop, K., Laimer, M., Nicolay, C., Reaney, M., Fink, K., Holl, R.W., 2016. Risk of severe hypoglycemia in sulfonylurea-treated patients from diabetes centers in Germany/Austria: how big is the problem? Which patients are at risk? Diabetes Metab. Res. Rev. 32, 316–324. https://doi.org/10.1002/dmrr.2722.
Sebranek, J.G., Sewalt, V.J.H., Robbins, K.L., Houser, T.A., 2005. Comparison of a natural rosemary extract and BHA/BHT for relative antioxidant effectiveness in pork sausage. Meat Sci. 69, 289–296. https://doi.org/10.1016/j.meatsci.2004.07.010.
Seven, A., Guzel, S., Seymen, O., Civelek, S., Bolayirli, M., Uncu, M., Barçak, G., 2004. Effects of vitamin E supplementation on oxidative stress in streptozotocin induced diabetic rats: investigation of liver and plasma. Yonsei Med. J. 45, 703–710. https://doi.org/10.3349/ymj.2004.45.4.703.
Sharma, P., Sharma, J.D., 2001. *In vitro* hemolysis of human erythrocytes-by plant extracts with antiplasmodial activity. J. Ethnopharmacol. 74 (3), 239–243. https://doi.org/10.1016/s0378-8741(00)00370-6.
Shieh, J.C.C., Huang, P.T., Lin, Y.F., 2020. Alzheimer's disease and diabetes: insulin signaling as the bridge linking two pathologies. Mol. Neurobiol. 57 (4), 966–1977. https://doi.org/10.1007/s12035-019-01858-5.
Shuo, Y., Xiao, Z., Shen, J., Li, J., Lai, H., Li, J., Chen, H., Zhao, Z., Yi, T., 2016. Rapid fingerprint analysis of Flos Carthami by ultra-performance liquid chromatography and similarity evaluation. JCS 54 (9), 1619–1624. https://doi.org/10.1093/chromsci/bmw115.
Šimić, G., Kostović, I., Winblad, B., Bogdanović, N., 1997. Volume and number of neurons of the human hippocampal formation in normal aging and Alzheimer's disease. J. Comp. Nerol. 379 (4), 482–494. https://doi.org/10.1002/(sici)1096-9861(19970324)379:4<482::aid-cne2>3.0.co;2-z.
Song, K., Kim, S., Na, J.Y., Park, J.H., Kim, J.K., Kim, J.H., Kwon, J., 2014. Rutin attenuates ethanol-induced neurotoxicity in hippocampal neuronal cells by increasing aldehyde dehydrogenase 2. FCT 72, 228–233. https://doi.org/10.1016/j.fct.2014.07.028.
Tang, Y.-N., He, X.-C., Ye, M., Huang, H., Chen, H.-L., Peng, W.-L., Zhao, Z.-Z., Yi, T., Chen, H.-B., 2015. Cardioprotective effect of total saponins from three medicinal

species of *Dioscorea* against isoprenaline-induced myocardial ischemia. J. Ethnopharmacol. 175, 451–455. https://doi.org/10.1016/j.jep.2015.10.004.

Tomlinson, D.R., Gardiner, N.J., 2008. Glucose neurotoxicity. Nat. Rev. Neurosci. 9 (1), 36–45. https://doi.org/10.1038/nrn2294.

Trifunović-Momčilov, M., Krstić-Milošević, D., Trifunović, S., Podolski-Renić, A., Pešić, M., Subotić, A., 2016. Secondary metabolite profile of transgenic centaury (Centaurium erythraea rafn.). Plants 1–26. https://doi.org/10.1007/978-3-319-27490-4_5-2.

Tsikas, D., 2017. Assessment of lipid peroxidation by measuring malondialdehyde (MDA) and relatives in biological samples: analytical and biological challenges. Anal. Biochem. 524, 13–30. https://doi.org/10.1016/j.ab.2016.10.021.

Uysal, S., Zengin, G., Mohomoodally, M.F., Yilmaz, M.A., Aktumsek, A., 2018. Chemical profile, antioxidant properties and enzyme inhibitory effects of the root extracts of selected *Potentilla* species. South Afr. J. Bot. 120, 124–128. https://doi.org/10.1016/j.sajb.2018.01.014.

Vargas-Restrepo, F., Sabogal-Guáqueta, A.M., Cardona-Gómez, G.P., 2018. Quercetin ameliorates inflammation in CA1 hippocampal region in aged triple transgenic Alzheimer's disease mice model. Biomedica 38, 1–23. https://doi.org/10.7705/biomedica.v38i0.3761.

Vital, P., Larrieta, E., Hiriart, M., 2006. Sexual dimorphism in insulin sensitivity and susceptibility to develop diabetes in rats. J. Endocrinol. 190 (2), 425–432. https://doi.org/10.1677/joe.1.06596.

von Gadow, A., Joubert, E., Hansmann, C.F., 1997. Comparison of the antioxidant activity of aspalathin with that of other plant phenols of rooibos tea (*Aspalathus linearis*), r-tocopherol, BHT, and BHA. J. Agric. Food Chem. 45 (3), 632–638. https://doi.org/10.1021/jf960281n.

Wang, C.P., Li, J.L., Zhang, L.Z., Zhang, X.C., Yu, S., Liang, X.M., Ding, F., Wang, Z.W., 2013. Isoquercetin protects cortical neurons from oxygen-glucose deprivation-reperfusion induced injury via suppression of TLR4-NF-kB signal pathway. Neurochem. Int. 63, 741–749. https://doi.org/10.1016/j.neuint.2013.09.018.

Wang, C.P., Shi, Y.W., Tang, M., Zhang, X.C., Gu, Y., Liang, X.M., Wang, Z.W., Ding, F., 2017. Isoquercetin ameliorates cerebral impairment in focal ischemia through anti-oxidative, anti-inflammatory, and anti-apoptotic effects in primary culture of rat hippocampal neurons and hippocampal CA1 region of rats. Mol. Neurobiol. 54, 2126–2142. https://doi.org/10.1007/s12035-016-9806-5.

Wang, L.F., Zhao, Y.Q., Gao, W.Z., Lou, J.S., Kang, Y., 2009. Dioscorea saponins increased antioxidative ability of myocardium after ischemia-perfusion injury in rat. Pharmacol. Clin. Chin. Mater. Med. 25 (5), 44–46.

Wang, Z., Yang, Y., Xiang, X., Zhu, Y., Men, J., He, M., 2010. Estimation of the normal range of blood glucose in rats. Wei sheng yan jiu= J.Hygiene Res. 39 (2), 133–137.

Wang, T.J., Choi, R.C., Li, J., Bi, C.W., Ran, W., Chen, X.H., Dong, T.T., Bi, K.S., Tsim, K.W., 2012. Trillin, a steroidal saponin isolated from the rhizomes of *Dioscorea nipponica*, exerts protective effects against hyperlipidemia and oxidative stress. J. Ethnopharmacol. 139, 214–220.

Wiater, A., Pleszczyńska, M., Próchniak, K., Szczodrak, J., 2008. Anticariogenic activity of the crude ethanolic extract of *Potentilla erecta* (L.) Raeusch. Herba Pol. J 54 (2), 41–45.

Wilkinson, P.D., Delgado, E.R., Alencastro, F., Leek, M.P., Roy, N., Weirich, M.P., Stahl, E.C., Otero, P.A., Chen, M.I., Brown, W.K., Duncan, A.W., 2019. The polyploid state restricts hepatocyte proliferation and liver regeneration in mice. Hepatology 69, 1242–1258. https://doi.org/10.1002/hep.30286.

Yan, Y., Zhou, Z., Kong, F., Feng, S., Li, X., Sha, Y., Zhang, G., Liu, H., Zhang, H., Wang, S., Hu, C., Zhang, X., 2016. Roux-en-Y gastric bypass surgery suppresses hepatic gluconeogenesis and increases intestinal gluconeogenesis in a T2DM rat model. Obes. Surg. 26, 2683–2690. https://doi.org/10.1007/s11695-016-2157-5.

Yang, B., Xie, Y., Guo, M., Rosner, M.H., Yang, H., Ronco, C., 2018. Nephrotoxicity and Chinese herbal medicine. Clin. J. Am. Soc. Nephrol. 13 (10), 1605–1611. https://doi.org/10.2215/CJN.11571017.

Yao, X., Zhu, L., Chen, Y., Tian, J., Wang, Y., 2013. In vivo and in vitro antioxidant activity and α-glucosidase, α-amylase inhibitory effects of flavonoids from *Chicorium gladulosum* seeds. Food Chem. 139 (1–4), 59–66. https://doi.org/10.1016/j.foodchem.2012.12.045.

Yehye, W.A., Rahman, N.A., Ariffin, A., Abd Hamid, S.B., Alhadi, A.A., Kadir, F.A., Yaeghoobi, M., 2015. Understanding the chemistry behind the antioxidant activities of butylated hydroxytoluene (BHT): a review. Eur. J. Med. Chem. https://doi.org/10.1016/j.ejmech.2015.06.026.

Yi, G.-L., Zhu, M.-Z., Cui, H.-C., Yuan, X.-R., Liu, P., Tang, J., Li, Y.-Q., Zhu, X.-H., 2022. A hippocampus dependent neural circuit loop underlying the generation of auditory mismatch negativity. Neuropharmacology 206, 108947. https://doi.org/10.1016/j.neuropharm.2022.108947.

Zeb, A., Haq, A., Murkovic, M., 2019. Effects of microwave cooking on carotenoids, phenolic compounds and antioxidant activity of *Cichorium intybus* L. (chicory) leaves. Eur. Food Res. Technol. 245, 365–374. https://doi.org/10.1007/s00217-018-3168-3.

Zeiger, E., 2003. Illusions of safety: antimutagens can be mutagens, and anticarcinogens can be carcinogens. Mutant. Res. 543, 191–194. https://doi.org/10.1016/S1383-5742(02)00111-4.

Zhang, L., He, S., Yang, F., Yu, H., Xie, W., Dai, Q., Zhang, D., Liu, X., Zhou, S., Zhang, K., 2016. Hyperoside ameliorates glomerulosclerosis in diabetic nephropathy by downregulating miR-21. Can. J. Physiol. Pharmacol. 94 (12), 1249–1256.

Zhang, R., Yao, Y., Wang, Y., Ren, G., 2011. Antidiabetic activity of isoquercetin in diabetic KK-A y mice. Nutr. Metab. 8 (85), 1743–7075.

Zhang, Q.W., Lin, L.G., Ye, W.C., 2018. Techniques for extraction and isolation of natural products: a comprehensive review. Chin. Med. 13, 1–26.

Zhang, Y., Wang, M., Dong, H., Yu, X., Zhang, J., 2018. Anti-hypoglycemic and hepatocyte-protective effects of hyperoside from *Zanthoxylum bungeanum* leaves in mice with high-carbohydrate/high-fat diet and alloxan-induced diabetes. Int. J. Mol. Med. 41, 77–86. https://doi.org/10.3892/ijmm.2017.3211.

Zhang, Y.W., Zhang, J.Q., Liu, C., Wei, P., Zhang, X., Yuan, Q.Y., Yin, X.T., Wei, L.Q., Cui, J.G., Wang, J., 2015. Memory dysfunction in type 2 diabetes mellitus correlates with reduced hippocampal CA1 and subiculum volumes. Chin. Med. J. 128, 465–471. https://doi.org/10.4103/0366-6999.151082.

Zhao, Y., Chen, B., Shen, J., Wan, L., Zhu, Y., Yi, T., Xiao, Z., 2017. The beneficial effects of quercetin, curcumin, and resveratrol in obesity. In: Oxidative Medicine and Cellular Longevity. Oxid Med Cell Longev, 1459497, 2017. https://www.hindawi.com/journals/omcl/2017/1459497/.

Zhou, J., Zhang, S., Sun, X., Lou, Y., Bao, J., Yu, J., 2021. Hyperoside ameliorates diabetic nephropathy induced by STZ via targeting the miR-499–5p/APC axis. J. Pharmacol. Sci. 146, 10–20. https://doi.org/10.1016/j.jphs.2021.02.005.

Životić, D., Životić, D., 1979. Lekovito Bilje U Narodnoj Medicini, fifth ed. Otokar keršovani – Rijeka, Belgrade. Yugoslavia.

Zlatković, B.K., Bogosavljević, S.S., Radivojević, A.R., Pavlović, M.A., 2013. Traditional use of the native medicinal plant resource of Mt. Rtanj (Eastern Serbia): ethnobotanical evaluation and comparison. J. Ethnopharmacol. 151, 704–713. https://doi.org/10.1016/j.jep.2013.11.037.